

02409 Multivariate Statistics

Lecture M, December 1 2025

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(1-3) 60%

Clustering 4 groups

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Groups

28

16

1

Factor 1 [41%]

Factor 3 [19%]

V. De Geneve

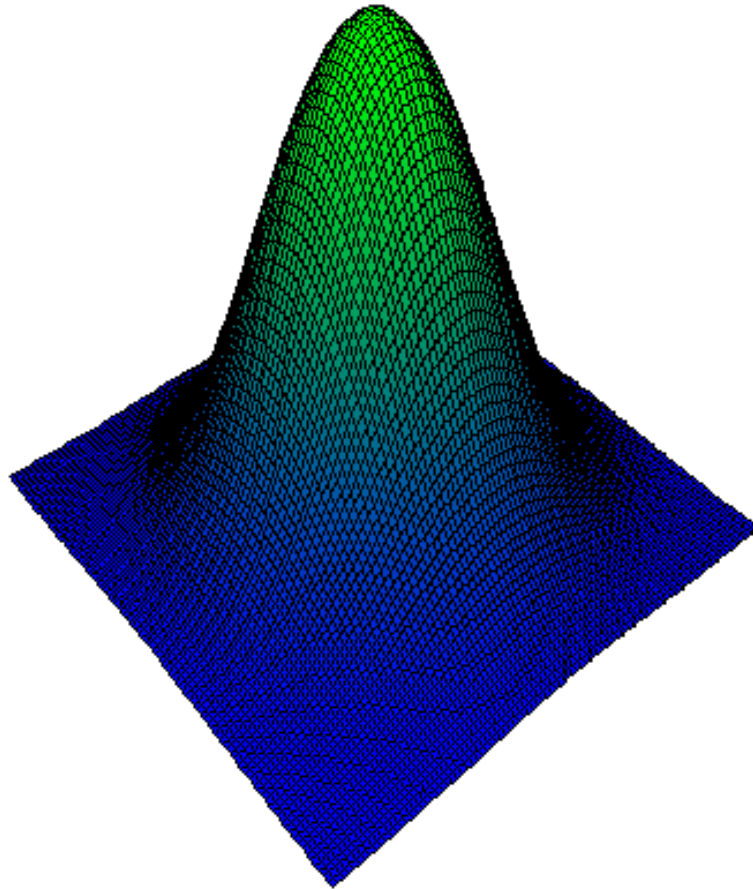
Agenda

- Review of curriculum
- Exam 2024

Course Topics

Topic	Book chapter	Lecture
The Multivariate Normal Distribution	1.1-1.6	A, B, C
Principal Components Analysis	6.1	C, D
Factor Analysis	6.3	D, E
Canonical Correlation Analysis	6.2	E
Discriminant Analysis	5	F, G
The General Linear Model	2.2, 3.1-3.4, 3.5.1-2.	G, H, I, J
Multivariate Statistical Analysis	4	J, K, L
Random Effects Models	2.3	L

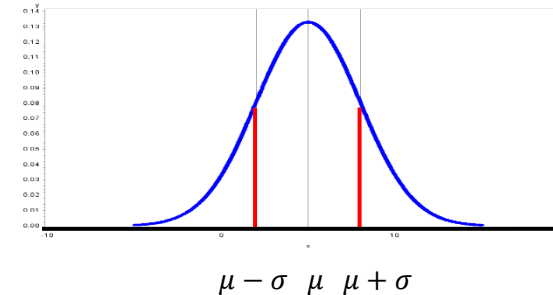
The Multivariate Normal Distribution



Recab: Uni- and bivariate normal distributions

The univariate random variable X is normally distributed $X \in N(\mu, \sigma^2)$ if

$$f(x) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} \exp\left(-\frac{1}{2\sigma^2} (x - \mu)^2\right)$$

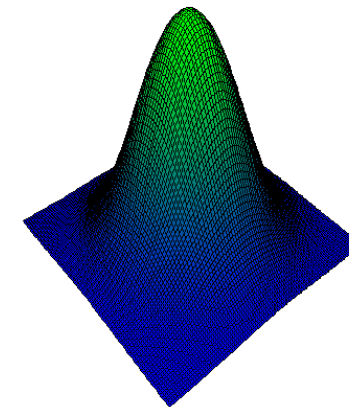


The bivariate random variable $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is normally distributed $\mathbf{X} \in N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ if the density is

$$f(\mathbf{x}) = \frac{1}{\sqrt{2\pi}^2} \frac{1}{\sqrt{\det \boldsymbol{\Sigma}}} \exp\left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

Example: $\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$; independent, identically distributed standard normal:

$$\begin{aligned} f(x_1, x_2) &= \frac{1}{2\pi} \exp\left(-\frac{1}{2} (x_1^2 + x_2^2)\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} x_1^2\right) \times \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} x_2^2\right) \end{aligned}$$



Independence in the multivariate normal distribution

||| Theorem 1.22

Let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right).$$

Then

$$X_i \sim N(\mu_i, \Sigma_{ii}),$$

and

$$X_1, X_2 \text{ are stochastically independent} \Leftrightarrow \Sigma_{12} = \Sigma_{21}^T = \mathbf{0},$$

where $\mathbf{0}$ is a null matrix.

Conditional regression

The variance $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ is *independent* of x_2 :

If (X_1, X_2) is multivariate normal, the residuals (ε_i) in the (potentially multivariate) regression

$$X_{1i} = \alpha + \beta x_{2i} + \varepsilon_i, i = 1, \dots, n$$

are independent and identically normal distributed. The converse is also true because

$$f_{XY}(x, y) = f_{X|Y=y}(x)f_Y(y)$$

THUS:

(X_1, X_2) is multivariate normal iff (X_1, X_2) is marginally normal, and the conditional distributions are normal!

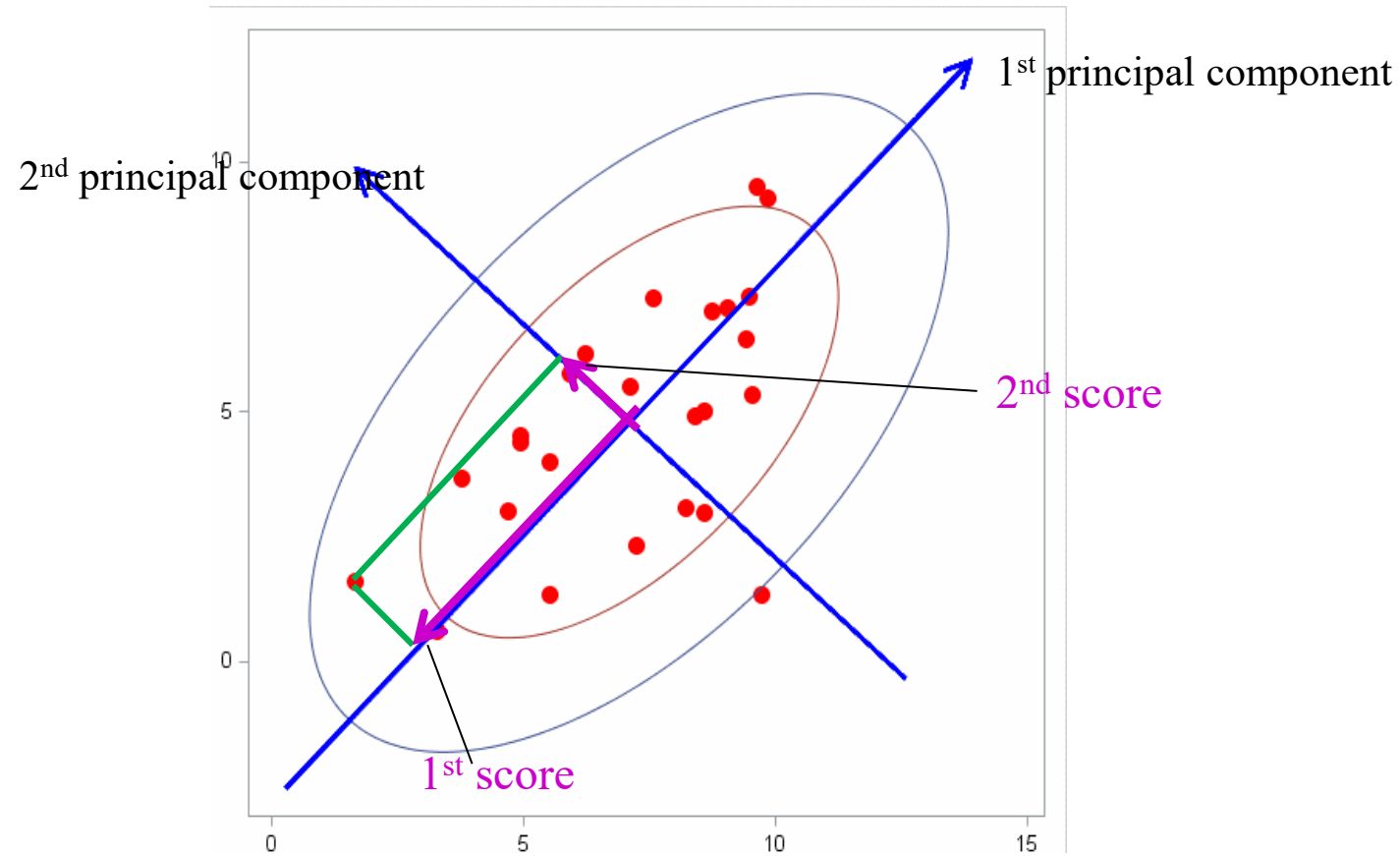
Principal Component Analysis

- PCA is a method to handle many variables, which are mutually correlated.
- PCA seeks to identify underlying dimensions in a data material, and to estimate the relationship between these.
- PCA may be used as a data reducing method, often in relation to multiple regression.
- PCA reduces the number of explanatory variables to a lesser number of "principal components", with (we hope) nearly as much of the variation as the initial variables.

What PCA does is essentially the contents of the preceding slides!

- The eigenvectors on the previous slides are **exactly** what is known as the **Principal Components**.

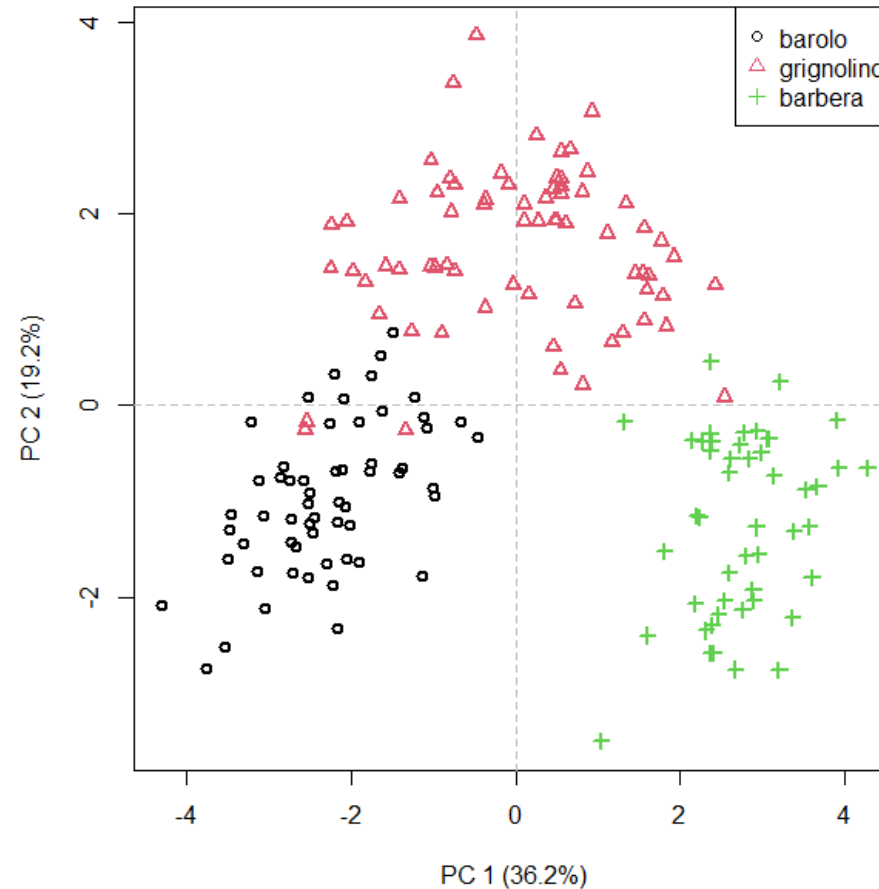
Principal Components I



The **scores** of the observation x_i of a multidimensional random variable X are the projections of x_i (adjusted for overall mean) on the principal components of the variance matrix of X

The Score plot

```
scoreplot(wines.PC, col = vintages, pch= as.numeric(vintages), lwd=2)  
legend("topright",levels(vintages), col=1:3,pch=1:3)
```

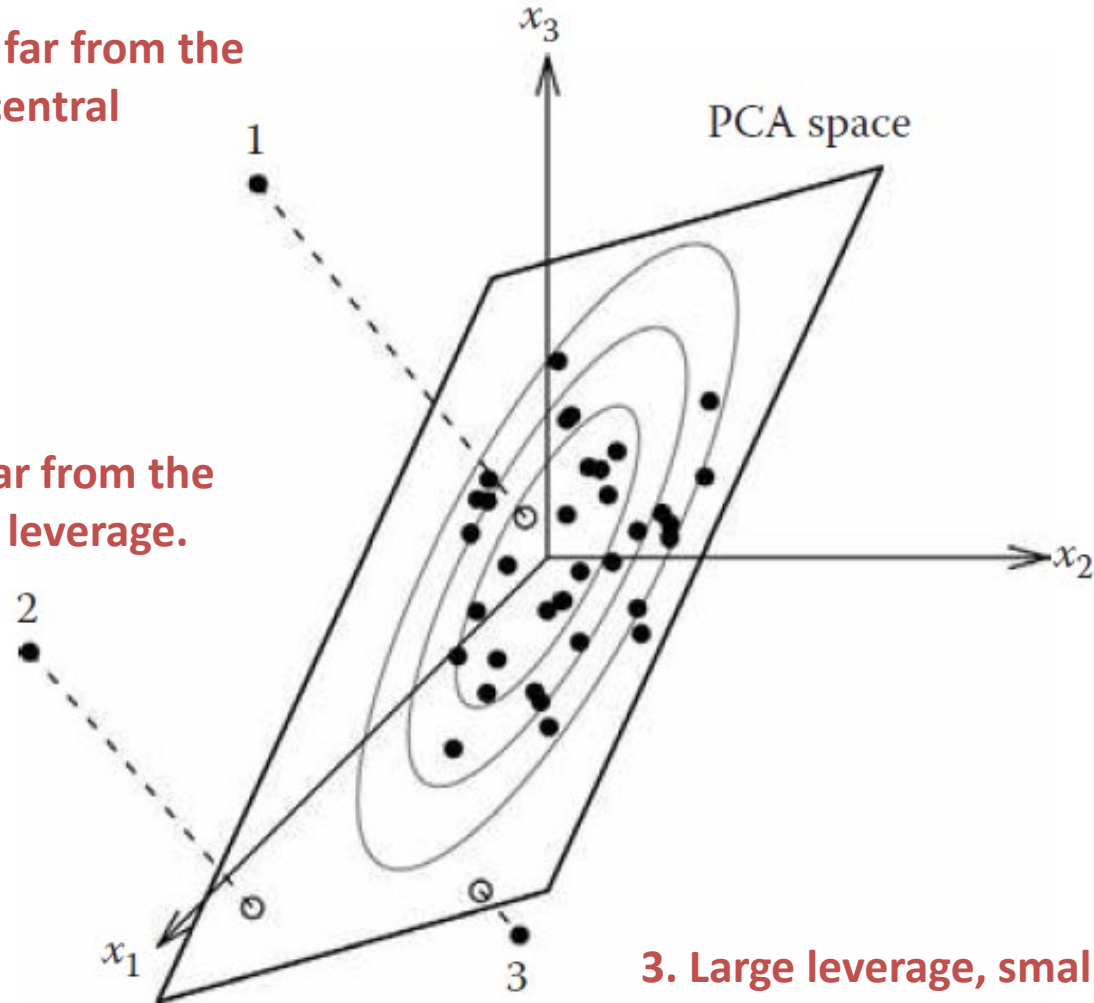


Apparently, we can
more or less separate
the vintages from just
TWO dimensions!

Diagnostic Plots – Residuals and Leverage

1. Large residual, far from the PCA space but central

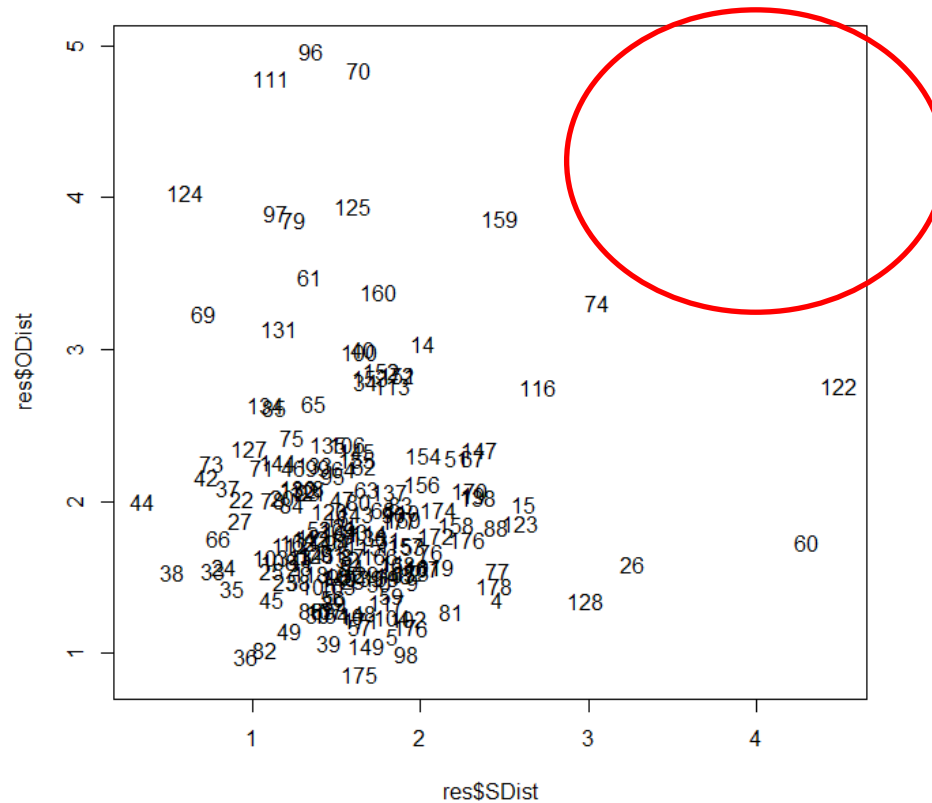
2. Large residual, far from the PCA space, large leverage.



3. Large leverage, small residual

Diagnostic Plots

```
par(mfrow=c(1,1))  
plot(res$SDist, res$ODist, type="n")  
text(res$SDist, res$ODist, labels=as.character(1:178))
```



No points have high leverage and also high residual

The Factor Model

$$[X_1 \quad \cdots \quad X_n] = A[F_1 \quad \cdots \quad F_n] + [G_1 \quad \cdots \quad G_n]$$

$$\mathbf{X} = \mathbf{A}\mathbf{F} + \mathbf{G}$$

$$\begin{bmatrix} X_1 \\ \vdots \\ X_k \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{km} \end{bmatrix} \begin{bmatrix} F_1 \\ \vdots \\ F_m \end{bmatrix} + \begin{bmatrix} G_1 \\ \vdots \\ G_k \end{bmatrix}$$

$$V(\mathbf{X}) = \mathbf{R} = \begin{bmatrix} 1 & \cdots & \rho_{1k} \\ \vdots & \ddots & \vdots \\ \rho_{k1} & \cdots & 1 \end{bmatrix}$$

$$V(\mathbf{F}) = \mathbf{I} = \mathbf{I}_m = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

$$V(\mathbf{G}) = \mathbf{\Delta} = \begin{bmatrix} \delta_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \delta_k \end{bmatrix}$$

$$\text{Cov}(\mathbf{F}, \mathbf{G}) = \mathbf{0} = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}$$

$\mathbf{X}$ ($X_1, \dots, X_n$) is (are) the observation(s),
 $\mathbf{A}$ an unknown matrix we want to determine,
 $\mathbf{F}$ ($F_1, \dots, F_n$) the unobservable common factors (**factor scores**), that we (perhaps) want to determine (aka **latent variables**), and
 $\mathbf{G}$ the unobservable unique factors

The factors are uncorrelated

Assumptions

The unique factors are uncorrelated

F and G are uncorrelated

The data has variance 1, i.e., the data is normalised / we use the correlation matrix

The Principal Factor Solution version 1

Let

$$\mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_k \end{bmatrix} \text{ with } V(\mathbf{X}) = \mathbf{R} = \begin{bmatrix} 1 & \cdots & \rho_{1k} \\ \vdots & \ddots & \vdots \\ \rho_{k1} & \cdots & 1 \end{bmatrix}, \quad \rho_{ij} = \rho_{ji}$$

Let the eigenvalues of $\mathbf{\Sigma} = \mathbf{R}$ be $\lambda_1 \geq \cdots \geq \lambda_k$ and the corresponding eigenvectors $\mathbf{p}_1, \cdots, \mathbf{p}_k$. Let $m \leq k$ and define

$$\mathbf{P}^{(m)} = [\mathbf{p}_1, \cdots, \mathbf{p}_m]$$

$$\mathbf{\Lambda}^{(m)} = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_m \end{bmatrix}$$

Then the **principal factor solution version 1** is

$$\mathbf{A} = [\mathbf{p}_1, \cdots, \mathbf{p}_m] \begin{bmatrix} \sqrt{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sqrt{\lambda_m} \end{bmatrix}$$

Remember from PCA:

$$\mathbf{R} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T = \mathbf{P}\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2}\mathbf{P}^T = \mathbf{P}\mathbf{\Lambda}^{1/2}(\mathbf{P}\mathbf{\Lambda}^{1/2})^T = \mathbf{A}\mathbf{A}^T$$

However, in general in the factor model,

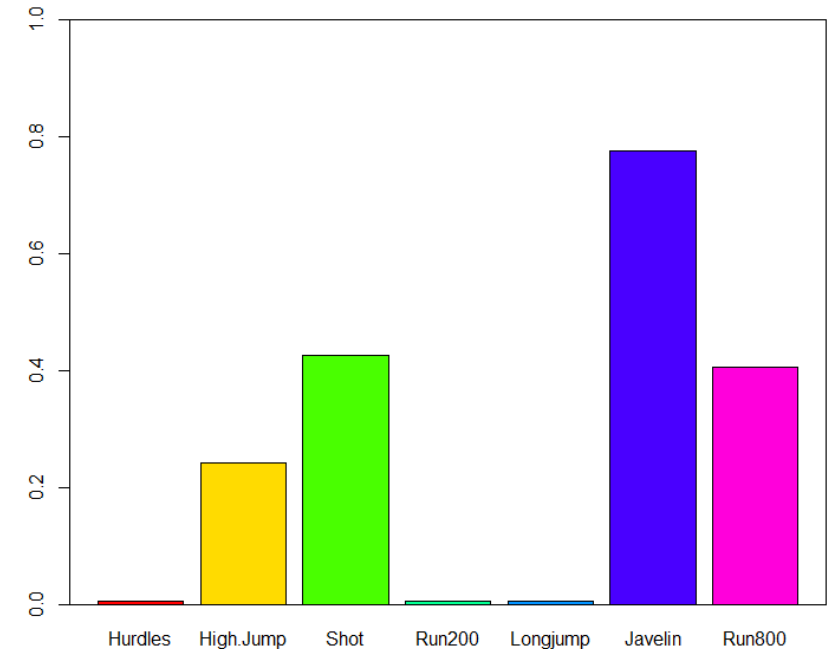
$$V(\mathbf{X}) = \mathbf{R} = V(\mathbf{A}\mathbf{F} + \mathbf{G}) = \mathbf{A}\mathbf{A}^T + \mathbf{\Delta}$$

Version 1 of the principal factor model **disregards contribution from the independent factor $\mathbf{G}$** .

Heptathlon data –uniqueness factors

```
barplot(temp$uniquenesses,col=rainbow(7),ylim=c(0,1))  
box()
```

- Hurdles, 200 Meter Run and Long Jump are essentially completely explained from the factors.
- High Jump, Shot Put, Jarvelin Throw and 800 Meter Run have a larger degree of unique variation, in particular Jarvelin Throw.
- High Jump, Shot Put and Jarvelin Throw are all disciplines with a technical element. 800 Meter Run has a tactical element where conservation of power is a main asset.
- Hurdles, 200 Meter Run and Long Jump, while also for Hurdles and Long Jump containing a technical element, are discipline where direct exposure of strength is a key element.



Canonical Correlation

y, x : **Correlation**. Data are divided into two variables x and y .

$y, x_1 + \dots + x_p$: **Multiple correlation**.

Maximal correlation between y and a linear combination of x . Data are divided into one variable y , and a set of variables $x_1, \dots, x_p$.

$y_1 + y_2 + \dots + y_p, x_1 + x_2 + \dots + x_q$: **Canonical correlation**.

Maximal correlation between a linear combination of y and a linear combination of x ; data are divided into sets of variables x and y .

The most "multivariate" of the correlation forms that we are going to study.

Canonical correlation identifies and quantifies the relations between *two sets of variables*.

Canonical correlation analysis identifies the *linear combination* of one set that is *maximally correlated* to a linear combination of the other set.

Canonical Correlation

$$\begin{aligned}(\Sigma_{yy}^{-1}\Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy} - \lambda_1\lambda_2)\mathbf{a} &= 0 \\(\Sigma_{xx}^{-1}\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx} - \lambda_1\lambda_2)\mathbf{b} &= 0\end{aligned}\tag{5}$$

Take

$$\begin{aligned}E_1 &= \Sigma_{yy}^{-1}\Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy} \\E_2 &= \Sigma_{xx}^{-1}\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}\end{aligned}$$

Note that E_1 is a $p \times p$ matrix, while E_2 is a $q \times q$ matrix.

Equation (5) then reads that $\mathbf{a}, \mathbf{b}$ are **eigenvectors** for respectively E_1, E_2 , with the **same** eigenvalue $\varrho^2 = \lambda_1\lambda_2$. This is optimized if ϱ^2 is taken as the largest eigenvalue for E_1 (and thus for E_2).

Obviously, this procedure may be iterated to find the maximal correlations perpendicular to $\mathbf{a}, \mathbf{b}$, respectively, as the remaining eigenvectors of E_1, E_2 .

Canonical Correlation

||| Theorem 6.12

Let the situation be given in the above mentioned definition and let $D(Z) = \Sigma$ be partitioned analogously

$$\Sigma = \begin{bmatrix} \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & \Sigma_{xx} \end{bmatrix}$$

Then the r' th canonical correlation is equal to the r' th largest root ϱ_r of

$$\det \begin{bmatrix} -\rho \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & -\rho \Sigma_{xx} \end{bmatrix} = 0$$

and the coefficients in the r' th pair of canonical variables satisfies

$$\begin{aligned} \text{(i)} \quad & \begin{bmatrix} -\varrho_r \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & -\varrho_r \Sigma_{xx} \end{bmatrix} \begin{bmatrix} a_r \\ b_r \end{bmatrix} = 0 \\ \text{(ii)} \quad & a_r^T \Sigma_{yy} a_r = 1 \\ \text{(iii)} \quad & b_r^T \Sigma_{xx} b_r = 1 \end{aligned}$$

||| Theorem 6.13

Let the situation be as in the previous theorem. Then we have

$$(\Sigma_{yx} \Sigma_{xx}^{-1} \Sigma_{xy} - \varrho_r^2 \Sigma_{yy}) a_r = 0$$

$$\det (\Sigma_{yx} \Sigma_{xx}^{-1} \Sigma_{xy} - \varrho_r^2 \Sigma_{yy}) = 0$$

respectively

$$(\Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx} - \varrho_r^2 \Sigma_{xx}) b_r = 0$$

$$\det (\Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx} - \varrho_r^2 \Sigma_{xx}) = 0$$

- Note that Theorem 6.12 concerns ϱ , and Theorem 6.13 concerns ϱ^2 . Finding the solutions through E_1, E_2 thus means that the canonical correlation ϱ_r is obtained as the square root of the eigenvalue ϱ_r^2 .

Canonical Correlation

Definition 6.11

Consider Z as above. Then the first *pair of canonical variables* is the pair of linear combinations

$$V_1 = a_1^T Y \text{ and } W_1 = b_1^T X$$

each having variance 1 that maximize the correlation $\rho(a^T Y, b^T X)$ for all (a, b) . The maximum correlation ρ_1 is the *first canonical correlation*. For $r \leq p$ we define the *r'th pair of canonical variables* as the pair of linear combinations

$$V_r = a_r^T Y \text{ and } W_r = b_r^T X ,$$

which each has the variance 1, which are uncorrelated with the previous $r - 1$ pairs of canonical variables, and which maximizes the correlation $\rho(a^T Y, b^T X)$ under those constraints. The maximum correlation ρ_r is the *r'th canonical correlation*.

Discrimination and Classification

1. The elements of discrimination and classification
2. Bayes and MINIMAX solutions
3. The Normal Case:
 1. Linear Discriminant Analysis
 2. Quadratic Discriminant Analysis
4. Mahalanobis' distance
5. Estimation of classification errors in LANDSAT data
6. Canonical Discriminant Analysis

Decision functions

- We seek a function to determine, based on observational data, if a subject belongs to π_1 or π_2 :

$$d: \mathbb{R}^p \rightarrow \{\pi_1, \pi_2\}$$

- Since d can only attain two values, it will necessarily subdivide $\mathbb{R}^p$ into two disjoint subsets R_1, R_2 :

$$d(x) = \begin{cases} \pi_1 & \text{if } x \in R_1 \\ \pi_2 & \text{if } x \in R_2 \end{cases}$$

- If we choose R_1 , we choose d !

Distribution of the Linear Discriminator

||| Theorem 5.8

We consider the random variable defined by the linear discriminator (omitting the term $-\log c$), i.e.

$$Z = X^T \Sigma^{-1} (\mu_1 - \mu_2) - \frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_2^T \Sigma^{-1} \mu_2 .$$

Then

$$Z \sim \begin{cases} \text{N} \left(+\frac{1}{2} \|\mu_1 - \mu_2\|_{\Sigma^{-1}}^2, \|\mu_1 - \mu_2\|_{\Sigma^{-1}}^2 \right) & \text{if } \pi_1 \text{ is true} \\ \text{N} \left(-\frac{1}{2} \|\mu_1 - \mu_2\|_{\Sigma^{-1}}^2, \|\mu_1 - \mu_2\|_{\Sigma^{-1}}^2 \right) & \text{if } \pi_2 \text{ is true} \end{cases} .$$

Example

$$\mu_1 = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \mu_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Sigma_1 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \Sigma_1^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \Sigma_2^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\frac{f_1(x)}{f_2(x)} \geq c \Leftrightarrow -(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) + (x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) \geq 2 \log(c)$$

$$\Leftrightarrow -\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \geq 2 \log(c)$$

$$\Leftrightarrow -x_1^2 + 2x_1x_2 + 10x_1 - 6x_2 - 18 \geq 2 \log(c)$$

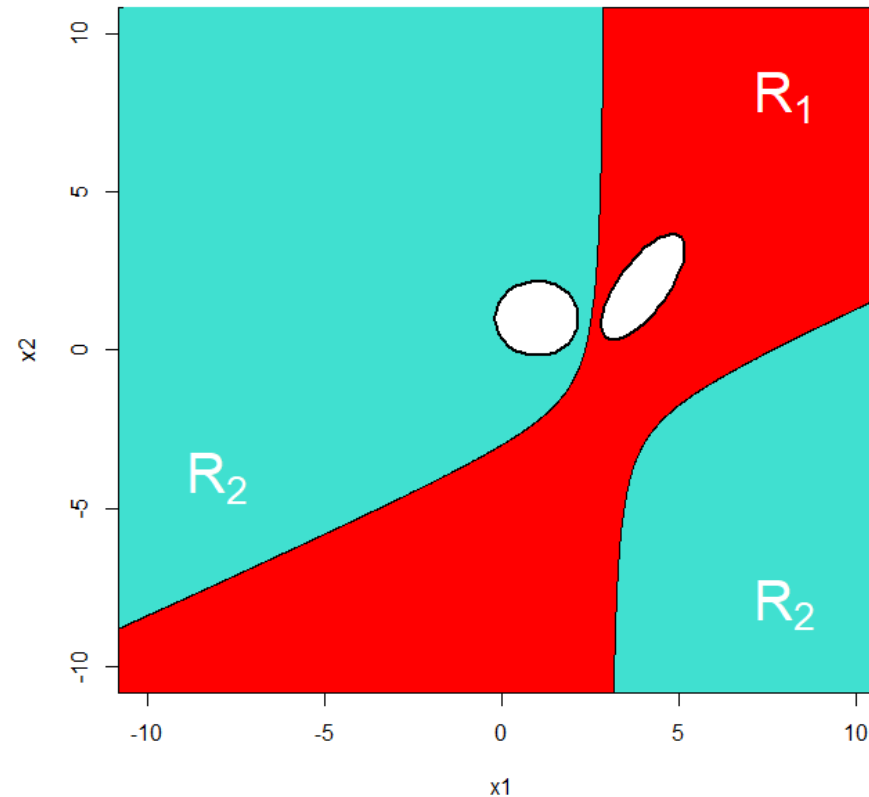
The discriminator is no longer linear in x!

Example

- Choose $c = 1$, equal loss

$$-x_1^2 + 2x_1x_2 + 10x_1 - 6x_2 - 18 \geq 0$$

- The separation of $\mathbb{R}^2$ into R_1 and R_2 is no longer a hyperplane, but the hyperbola given above:



The General Linear Model - GLM

$$Y \sim N_n(\mu, \sigma^2 \Sigma)$$

$$\mu = X\theta$$

$$\begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = \begin{bmatrix} x_{11} & \cdots & x_{1k} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_k \end{bmatrix}$$

X is the **design matrix**; assumed known, and with full rank

Σ is assumed known

σ^2 is assumed unknown

The General Linear Model - GLM

||| Theorem 2.3

Let $\mathbf{x}$ and $\boldsymbol{\theta}$ be given as in the preceding section and let $\mathbf{Y} \sim N_n(\mathbf{x}\boldsymbol{\theta}, \sigma^2\boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is positive definite. Then the maximum likelihood estimator $\hat{\boldsymbol{\theta}}$ for $\boldsymbol{\theta}$ is given by $\mathbf{x}\hat{\boldsymbol{\theta}}$ being the projection (with respect to $\boldsymbol{\Sigma}$) onto M , $\hat{\boldsymbol{\theta}}$ is a solution to the so-called *normal equation(s)*

$$(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x})\hat{\boldsymbol{\theta}} = \mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{y}.$$

If $\mathbf{x}$ has full rank k , then

$$\hat{\boldsymbol{\theta}} = (\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x})^{-1}\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{Y},$$

and being a linear combination of normally distributed variables $\hat{\boldsymbol{\theta}}$ is also normally distributed with parameters

$$\begin{aligned} E(\hat{\boldsymbol{\theta}}) &= \boldsymbol{\theta} \\ D(\hat{\boldsymbol{\theta}}) &= \sigma^2(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x})^{-1}. \end{aligned}$$

It is especially noted that $\hat{\boldsymbol{\theta}}$ is an unbiased estimate of $\boldsymbol{\theta}$.

The General Linear Model - GLM

||| Theorem 2.5

Let the situation be as above. The maximum likelihood estimator of σ^2 is

$$\tilde{\sigma}^2 = \frac{1}{n} \|Y - \mathbf{x} \hat{\theta}\|^2 = \frac{1}{n} (Y - \mathbf{x} \hat{\theta})^T \Sigma^{-1} (Y - \mathbf{x} \hat{\theta}).$$

The unbiased estimator of σ^2 is

$$\begin{aligned} \hat{\sigma}^2 &= \frac{1}{n - \text{rk}(\mathbf{x})} \|Y - \mathbf{x} \hat{\theta}\|^2 \\ &= \frac{1}{n - \text{rk}(\mathbf{x})} (Y - \mathbf{x} \hat{\theta})^T \Sigma^{-1} (Y - \mathbf{x} \hat{\theta}) \end{aligned}$$

where $\mathbf{x} \hat{\theta}$ is the maximum likelihood estimator of $E(Y)$. The following holds

$$\hat{\sigma}^2 \sim \sigma^2 \chi^2(n - \text{rk}(\mathbf{x})) / (n - \text{rk}(\mathbf{x}))$$

and $\hat{\sigma}^2$ is independent of the maximum likelihood estimator of the expected value and is therefore independent of $\hat{\theta}$.

GLM in R

- The `lm()` function.

```
synprod <- data.frame("Acontent"=c(1, 0, 0.5),  
                      "Bcontent"=c(0, 1, 0.5),  
                      "Qual"=c(90, 30, 75))
```

```
glmSyn <- lm(Qual ~ Acontent + Bcontent - 1, data=synprod)  
# the "-1" in the line above removes the intercept  
# If not specified an intercept is estimated by default
```

Hotelling's T-Square in the Two-Sample Case

||| Theorem 4.9

We use the same notation as given above. Now, let

$$T^2 = \frac{nm}{n+m}(\bar{X} - \bar{Y})^T S^{-1}(\bar{X} - \bar{Y}).$$

Then the critical region for a test of H_0 against H_1 at level α is equal to

$$C = \{x_1, \dots, x_n, y_1, \dots, y_m \mid \frac{n+m-p-1}{(n+m-2)p} t^2 > F(p, n+m-p-1)_{1-\alpha}\}$$

Here t^2 is the observed value of T^2 .

Likelihood Estimation

In the Multivariate Normal Model

- Likelihood function:

$$L(\theta, \Sigma) = \frac{1}{(2\pi)^{np/2}} \det(\Sigma)^{-n/2} \exp \left(-\frac{1}{2} \|Y - \theta\|_{I_n \otimes \Sigma^{-1}}^2 \right)$$

- We need to project Y wrt. the inner product induced by $I_n \otimes \Sigma^{-1}$ to obtain the maximum likelihood estimator.

Likelihood Estimation

In the Multivariate Normal Model

||| Theorem 4.14

We consider the above mentioned situation. If the observations Y_i are normally distributed the maximum likelihood estimate of θ is given by

$$\hat{\theta} = (x^T x)^{-1} x^T Y.$$

||| Theorem 4.19

We consider the situation from theorems 4.14 and 4.18 and we introduce the usual notations

$$\begin{aligned}\tilde{\theta} &= \text{vc}(\theta) = \begin{bmatrix} \theta_{|1} \\ \vdots \\ \theta_{|p} \end{bmatrix} \\ \hat{\hat{\theta}} &= \text{vc}(\hat{\theta}) = \begin{bmatrix} \hat{\theta}_{|1} \\ \vdots \\ \hat{\theta}_{|p} \end{bmatrix}.\end{aligned}$$

Then we have that $\hat{\hat{\theta}}$ is normally distributed

$$\hat{\hat{\theta}} = \text{vc}(\hat{\theta}) \sim N_{pk}(\tilde{\theta}, \Sigma \otimes (x^T x)^{-1}),$$

and $n\hat{\Sigma}^*$ is Wishart distributed

$$n\hat{\Sigma}^* \sim W(n - k, \Sigma).$$

Finally $\hat{\Sigma}^*$ and $\hat{\hat{\theta}}$ and therefore also $\hat{\Sigma}^*$ and $\hat{\theta}$ are stochastically independent.

- Note that the book (here) considers the observations as stacked in a columns and not as rows in a matrix. Hence the change in tensors...

Testing in the Multivariate Normal Model

$$H_1: \mu \in L_1 \otimes \mathbb{R}^p, \dim(L_1) = d_1$$

We note that

$$I_n - P_1 = (I - P_0) + (P_0 - P_1)$$

Where

$$(I - P_0)(P_0 - P_1) = 0$$

because $L_1 \otimes \mathbb{R}^p \subseteq L_0 \otimes \mathbb{R}^p$, so that with $SSD_1 = Y^T(P_0 - P_1)Y$ it holds that SSD_0 and SSD_1 are independent, and

$$SSD_{01} = SSD_0 + SSD_1$$

and thus

$$R_1 = Q^{2/n} = \frac{\det(SSD_0)}{\det(SSD_0 + SSD_1)}$$

Testing in the Multivariate Normal Model

- Take $A = Q_{01}^T X$, where the columns in Q_{01} form an orthonormal basis for $L_0 \cap L_1^\perp$, $B = I_p$ $C = 0$:
- $E = SSD_0$
- $\Delta = Q_{01}^T X \hat{\theta} = Q_{01}^T P_0 Y$
 $= Q_{01}^T Y$
- $H = Y^T Q_{01} (Q_{01}^T X (X^T X)^{-1} X^T Q_{01})^{-1} Q_{01}^T Y$
 $= Y^T Q_{01} I_{d_0 - d_1} Q_{01}^T Y$
 $= Y^T (P_0 - P_1) Y$
 $= SSD_1$

||| Theorem 4.21

We consider the above mentioned situation including the assumption of the normality of the observations. Furthermore we consider the hypothesis

$$H_0 : A \theta B^T = C \quad \text{against} \quad H_1 : A \theta B^T \neq C,$$

where $A(r \times k)$, $B(s \times p)$ and $C(r \times s)$ are given (known) matrices. We introduce

$$\Delta = A \hat{\theta} B^T - C$$

$$R = n \hat{\Sigma}^* = (Y - X \hat{\theta})^T (Y - X \hat{\theta}) = Y^T Y - \hat{\theta}^T (X^T X) \hat{\theta}$$

and

$$E = B R B^T$$

$$H = \Delta^T [A (X^T X)^{-1} A^T]^{-1} \Delta.$$

The likelihood ratio test for testing H_0 against H_1 is equivalent to the test given by the critical region

$$\{y \mid \frac{\det(e)}{\det(e + h)} \leq U(s, r, n - k)_\alpha\},$$

where $U(s, r, n - k)_\alpha$ is the α quantile in the null-hypothesis distribution of the test statistic (see below).

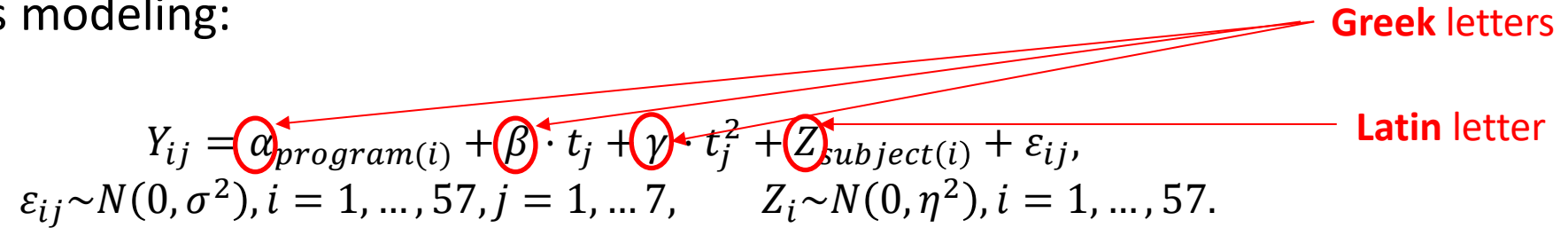
Random Effects Models

Example: Strength data

- Random effects modeling:

$$Y_{ij} = \alpha_{\text{program}(i)} + \beta \cdot t_j + \gamma \cdot t_j^2 + Z_{\text{subject}(i)} + \varepsilon_{ij},$$

$\varepsilon_{ij} \sim N(0, \sigma^2), i = 1, \dots, 57, j = 1, \dots, 7, \quad Z_i \sim N(0, \eta^2), i = 1, \dots, 57.$



- We **RANDOMIZE** the effect of *subject*; We move it from the systematic part of the model to the random part of the model.
- Gain: A prospective model valid for all (also future) subjects;
- Loss: Increased uncertainty.
- Loss: Introduction of dependence within subjects.

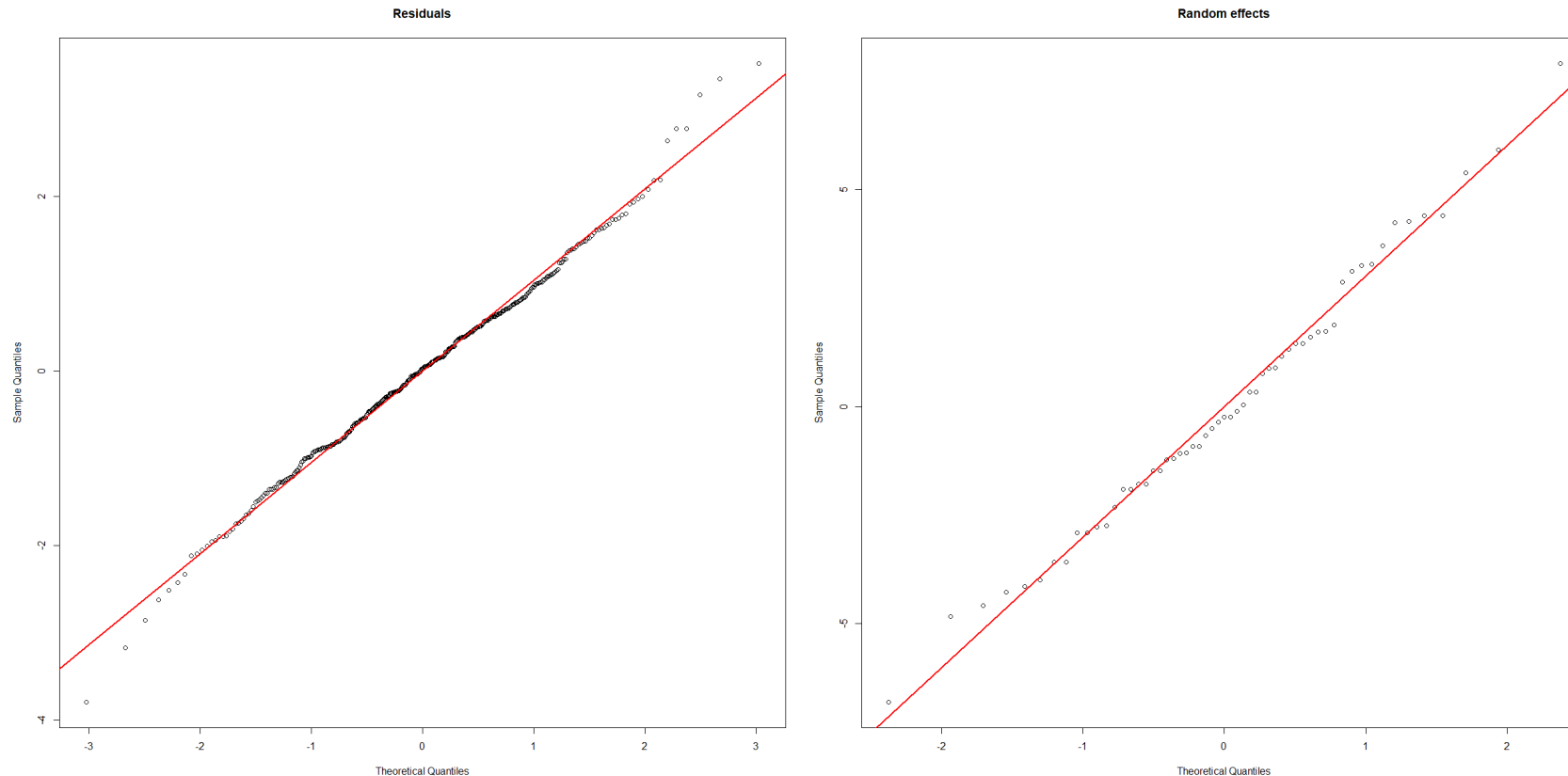
Random Effects Models

Example: Strength data

Implementation in **R**: `Model control`.

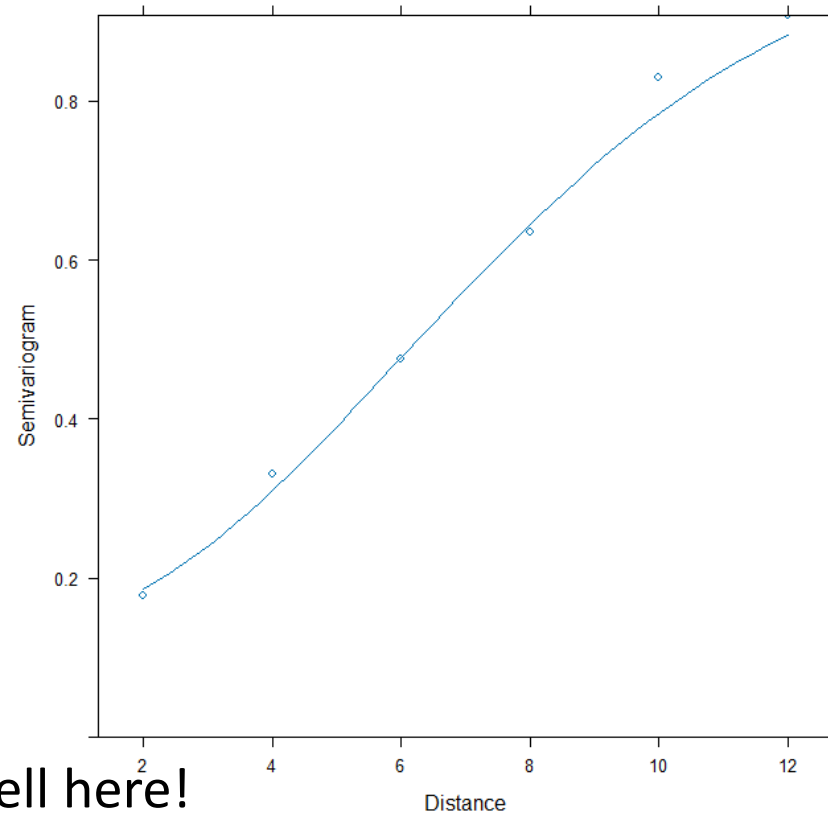
Residuals ($\hat{\epsilon}$): `residuals(model1)`.

Estimated random effects ($\hat{Z}$): `ranef(model1)`.



Random Effects Models: Advanced Correlation Structures The Semivariogram

- Semivariogram with Gaussian decay:



- Theory and practice fits well here!
- We select the model with a Spatial Gaussian Covariance structure.

Exam 2024

02409 MultivariateStatistics - 2024

4 hours written exam - All aids allowed, including internet access.

There are 5 problems with a total of 30 questions.

The questions in a specific problem should be read in order.

The questions are weighted equally

A correct answer gives 5 points, a wrong answer gives -1 point. Unanswered questions or a “don’t know” give 0 points. The total number of points needed for a satisfactorily answered exam is determined at the final evaluation of the exam. Especially note, that the grade 10 may be given even if only one answer is wrong or unanswered.

If data is needed to solve a specific problem, it will be available in the problem description as an R and a SAS file.

The problem will specifically say, if you need to use the dataset.

R hint: rds files can be loaded as follows: `df <- readRDS('data.rds')`

Problem 1

PROBLEM 1 - Room Occupancy

We consider the dataset from this source: <https://archive.ics.uci.edu/dataset/864/room+occupancy+estimation>

The data we will use is available here in R (<https://designer.mcq.eksamen.dtu.dk/api/images/7b747e85-9052-4934-a6c8-74dfc605d75e>) and in SAS (<https://designer.mcq.eksamen.dtu.dk/api/images/69102c55-82f4-419b-a221-3da407ed532b>) .

The study aims at predicting room occupancy by means of a number of variables: Sound, temperature, CO2, etc.

We will only consider the sound and temperature variables, and thus use a subset of the variables

S1_Temp S2_Temp S3_Temp S4_Temp

S1_Sound S2_Sound S3_Sound S4_Sound

In this problem we will ignore the outcome *RoomCount*, i.e., how many people are present in the room.

```
> occupancy<-readRDS('occupancy.rds')
```

```
> summary(occupancy)
```

S1_Temp	S2_Temp	S3_Temp	S4_Temp	S1_Light	S2_Light	S3_Light	S4_Light	S1_Sound	S2_Sound
Min. :24.94	Min. :24.75	Min. :24.44	Min. :24.94	Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. :0.0600	Min. :0.0400
1st Qu.:25.19	1st Qu.:25.19	1st Qu.:24.69	1st Qu.:25.44	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.:0.0700	1st Qu.:0.0500
Median :25.38	Median :25.38	Median :24.94	Median :25.75	Median : 0.00	Median : 0.00	Median : 0.00	Median : 0.00	Median :0.0800	Median :0.0500
Mean :25.45	Mean :25.55	Mean :25.06	Mean :25.75	Mean : 25.45	Mean : 26.02	Mean : 34.25	Mean :13.22	Mean :0.1682	Mean :0.1201
3rd Qu.:25.63	3rd Qu.:25.63	3rd Qu.:25.38	3rd Qu.:26.00	3rd Qu.: 12.00	3rd Qu.: 14.00	3rd Qu.: 50.00	3rd Qu.:22.00	3rd Qu.:0.0800	3rd Qu.:0.0600
Max. :26.38	Max. :29.00	Max. :26.19	Max. :26.56	Max. :165.00	Max. :258.00	Max. :280.00	Max. :74.00	Max. :3.8800	Max. :3.4400

S3_Sound	S4_Sound	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR	RoomCount
Min. :0.0400	Min. :0.0500	Min. : 345.0	Min. : -6.29615	Min. :0.00000	Min. :0.00000	Min. :0.0000
1st Qu.:0.0600	1st Qu.:0.0600	1st Qu.: 355.0	1st Qu.: -0.04615	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000
Median :0.0600	Median :0.0800	Median : 360.0	Median : 0.00000	Median :0.00000	Median :0.00000	Median :0.0000
Mean :0.1581	Mean :0.1038	Mean : 460.9	Mean : -0.00483	Mean :0.09014	Mean :0.07957	Mean :0.3986
3rd Qu.:0.0700	3rd Qu.:0.1000	3rd Qu.: 465.0	3rd Qu.: 0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.0000
Max. :3.6700	Max. :3.4000	Max. :1270.0	Max. : 8.98077	Max. :1.00000	Max. :1.00000	Max. :3.0000

Question 1.1

Question 1.1

The covariance between S1_Temp and S3_Temp is:

Choose one answer

- ☐ 0.1424
- ☐ 0.0146
- ☐ 0.1234
- ☐ 0.7997
- ☐ 0.5863
- ☐ Don't know

```
> cov(occupancy$S1_Temp, occupancy$S3_Temp)
[1] 0.1424453
```

- 0.1424 - option 1

Question 1.2

Question 1.2

The 80% confidence interval for the *correlation* between S1_Sound and S2_Sound is:

Choose one answer

- ☐ [0.551; 0.569]
- ☐ [0.547; 0.573]
- ☐ Don't know
- ☐ [0.511; 0.548]
- ☐ [0.542; 0.577]
- ☐ [0.561; 0.595]

||| Theorem 1.37

Let $R = R_{ij|m+1\dots p}$ be the empirical partial correlation coefficient between Z_i and Z_j conditioned on (or: for given) $Z_{m+1}, \dots, Z_p$. It is assumed to be computed from the unbiased estimates of the variance-covariance matrix and from n observations. Then

$$\frac{R}{\sqrt{1-R^2}} \sqrt{n-2-(p-m)} \sim t(n-2-(p-m)),$$

if $\rho_{ij|m+1,\dots,p} = 0$.

Question 1.2

||| Theorem 1.37

Let $R = R_{ij|m+1\dots p}$ be the empirical partial correlation coefficient between Z_i and Z_j conditioned on (or: for given) $Z_{m+1}, \dots, Z_p$. It is assumed to be computed from the unbiased estimates of the variance-covariance matrix and from n observations. Then

$$\frac{R}{\sqrt{1-R^2}} \sqrt{n-2-(p-m)} \sim t(n-2-(p-m)),$$

if $\rho_{ij|m+1\dots p} = 0$.

$$\frac{R}{\sqrt{1-R^2}} \sim \frac{t(n-2)}{\sqrt{n-2}} : ci_{80} = \left[\frac{-q_{0.9}(n-2)}{\sqrt{n-2}}; \frac{q_{0.9}(n-2)}{\sqrt{n-2}} \right] = [-c; c]$$

```
> (n<-dim(occupancy)[1])
[1] 10129
> (my.c<-qt(0.9,df=n-2)/sqrt(n-2))
[1] 0.01273573
```

Question 1.2

||| Theorem 1.37

Let $R = R_{ij|m+1\dots p}$ be the empirical partial correlation coefficient between Z_i and Z_j conditioned on (or: for given) $Z_{m+1}, \dots, Z_p$. It is assumed to be computed from the unbiased estimates of the variance-covariance matrix and from n observations. Then

$$\frac{R}{\sqrt{1-R^2}} \sqrt{n-2-(p-m)} \sim t(n-2-(p-m)),$$

if $\rho_{ij|m+1,\dots,p} = 0$.

$$\frac{R}{\sqrt{1-R^2}} \sim \frac{t(n-2)}{\sqrt{n-2}} : f(R) = \frac{R}{\sqrt{1-R^2}}; g(y) = f^{-1}(y) = \sqrt{\frac{y^2}{1+y^2}}$$

Thus with true $R = 0$:

$$0.9 = P(\hat{R} \in (R - g(c); R + g(c))) = P(R \in (\hat{R} - g(c); \hat{R} + g(c)))$$

Confidence interval: $(\hat{R} - g(c); \hat{R} + g(c))$

Coding:

```
> cor(occupancy$S1_Sound, occupancy$S2_Sound) - sqrt(my.c^2 / (1+my.c^2))  
[1] 0.5473269  
> cor(occupancy$S1_Sound, occupancy$S2_Sound) + sqrt(my.c^2 / (1+my.c^2))  
[1] 0.5727963
```

- Option 2- [0.547;0.573]

Question 1.3

Question 1.3

Assuming the correlation stays unchanged, the minimum number of samples needed for the correlation between S1_Sound & S2_Sound - when tested against 0 - to have a p-value of less than 0.1 is:

Choose one answer

- ☐ 15
- ☐ 13
- ☐ 10
- ☐ 6
- ☐ 8
- ☐ Don't know

Transforming into $\frac{R}{\sqrt{1-R^2}}; \quad \frac{R}{\sqrt{1-R^2}}\sqrt{n-2} \sim t(n-2)$:

```
(my.T<-cor(occupancy$S1_Sound,occupancy$S2_Sound) /  
sqrt(1+cor(occupancy$S1_Sound,occupancy$S2_Sound)^2))  
[1] 0.4886443
```

Question 1.3

Question 1.3

Assuming the correlation stays unchanged, the minimum number of samples needed for the correlation between S1_Sound & S2_Sound - when tested against 0 - to have a p-value of less than 0.1 is:

Choose one answer

- ☐ 15
- ☐ 13
- ☐ 10
- ☐ 6
- ☐ 8
- ☐ Don't know

Comparing lhs and rhs for the options for n (and 16); $p < 0.1$ means exceeding a 2-sided ci_{90} interval:

$$\frac{R}{\sqrt{1-R^2}}\sqrt{n-2} \sim t(n-2)$$

```
> my.T*sqrt(c(6,8,10,13,15,16)-2)
[1] 0.9772886 1.1969292 1.3820947 1.6206497 1.7618320 1.8283395
> qt(0.95,df=c(6,8,10,13,15,16)-2)
[1] 2.131847 1.943180 1.859548 1.795885 1.770933 1.761310
```

Actually, the answer is 16...

Probably meant a one sided test:

```
> qt(0.90,df=c(6,8,10,13,15,16)-2)
[1] 1.533206 1.439756 1.396815 1.363430 1.350171 1.345030
```

which gives 13 as $1.6206497 > 1.363430$, as the first sample size; option 2.

Question 1.4

Question 1.4

Using a linear combination of the temperature variables, i.e. with no intercept, the fraction of variance explained of S1_Sound is:

Choose one answer

- ☐ Don't know
- ☐ 0.138
- ☐ 0.301
- ☐ 0.229
- ☐ 0.549
- ☐ 0.278

- The level of variance explained is the R^2 statistic, which we read off directly from the `lm` summary, coded with a -1:

```
> summary(lm(S1_Sound~-1+S1_Temp+S2_Temp+S3_Temp+S4_Temp,data=occupancy))$r.square  
[1] 0.3013477
```

- $R^2 = 0.301$, option 3.

Question 1.5

Question 1.5

We now assume that using a linear combination of the temperature variables, the fraction of variance explained of S4_Sound is 0.1530. When testing this value against 0, the resulting F-value is:

Choose one answer

- ☐ Don't know
- ☐ 367.42
- ☐ 457.19
- ☐ 585.36
- ☐ 418.27
- ☐ 113.83

||| Theorem 1.45

Let $R = \hat{\rho}_{y_i|x}$ be the empirical multiple correlation coefficient between Y_i and $\mathbf{X} = (Z_{m+1}, \dots, Z_p)$ based upon n observations. Then

$$\frac{R^2}{1 - R^2} \cdot \frac{n - (p - m) - 1}{p - m} \sim F(p - m, n - (p - m) - 1),$$

if $\rho_{y_i|x} = \rho_{y_i|z_{m+1}, \dots, z_p} = 0$.

Question 1.5

Question 1.5

We now assume that using a linear combination of the temperature variables, the fraction of variance explained of S4_Sound is 0.1530. When testing this value against 0, the resulting F-value is:

Choose one answer

- ☐ Don't know
- ☐ 367.42
- ☐ 457.19
- ☐ 585.36
- ☐ 418.27
- ☐ 113.83

||| Theorem 1.45

Let $R = \hat{\rho}_{y_i|x}$ be the empirical multiple correlation coefficient between Y_i and $X = (Z_{m+1}, \dots, Z_p)$ based upon n observations. Then

$$\frac{R^2}{1 - R^2} \cdot \frac{n - (p - m) - 1}{p - m} \sim F(p - m, n - (p - m) - 1),$$

if $\rho_{y_i|x} = \rho_{y_i|z_{m+1}, \dots, z_p} = 0$.

- With $R^2 = 0.1530$, $n=10129$ and 4 temperature variables, ie. $p - m = 4$, we construct F as

```
> (0.1530 / (1 - 0.1530)) * (10129 - 5) / 4  
[1] 457.1936
```

- $F = 457.19$, option 3.

Question 1.6

Question 1.6

Using a linear combination of the temperature variables, the maximum fraction of variance that can be explained of a linear combination of the sound variables is:

Choose one answer

- ☐ Don't know
- ☐ 0.450
- ☐ 0.302
- ☐ 0.961
- ☐ 0.010
- ☐ 0.403

- Answer: the squared (first) canonical correlation between temperature and sound!

Lecture E - Canonical Correlation

$$\begin{aligned}(\Sigma_{yy}^{-1}\Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy} - \lambda_1\lambda_2)\mathbf{a} &= 0 \\(\Sigma_{xx}^{-1}\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx} - \lambda_1\lambda_2)\mathbf{b} &= 0\end{aligned}\tag{5}$$

Take

$$\begin{aligned}E_1 &= \Sigma_{yy}^{-1}\Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy} \\E_2 &= \Sigma_{xx}^{-1}\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}\end{aligned}$$

Note that E_1 is a $p \times p$ matrix, while E_2 is a $q \times q$ matrix.

Equation (5) then reads that $\mathbf{a}, \mathbf{b}$ are **eigenvectors** for respectively E_1, E_2 , with the **same** eigenvalue $\varrho^2 = \lambda_1\lambda_2$. This is optimized if ϱ^2 is taken as the largest eigenvalue for E_1 (and thus for E_2).

Obviously, this procedure may be iterated to find the maximal correlations perpendicular to $\mathbf{a}, \mathbf{b}$, respectively, as the remaining eigenvectors of E_1, E_2 .

Lecture E - Canonical Correlation

||| Theorem 6.12

Let the situation be given in the above mentioned definition and let $D(Z) = \Sigma$ be partitioned analogously

$$\Sigma = \begin{bmatrix} \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & \Sigma_{xx} \end{bmatrix}$$

Then the r' th canonical correlation is equal to the r' th largest root ϱ_r of

$$\det \begin{bmatrix} -\rho \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & -\rho \Sigma_{xx} \end{bmatrix} = 0$$

and the coefficients in the r' th pair of canonical variables satisfies

$$\begin{aligned} \text{(i)} \quad & \begin{bmatrix} -\varrho_r \Sigma_{yy} & \Sigma_{yx} \\ \Sigma_{xy} & -\varrho_r \Sigma_{xx} \end{bmatrix} \begin{bmatrix} a_r \\ b_r \end{bmatrix} = 0 \\ \text{(ii)} \quad & a_r^T \Sigma_{yy} a_r = 1 \\ \text{(iii)} \quad & b_r^T \Sigma_{xx} b_r = 1 \end{aligned}$$

||| Theorem 6.13

Let the situation be as in the previous theorem. Then we have

$$(\Sigma_{yx} \Sigma_{xx}^{-1} \Sigma_{xy} - \varrho_r^2 \Sigma_{yy}) a_r = 0$$

$$\det (\Sigma_{yx} \Sigma_{xx}^{-1} \Sigma_{xy} - \varrho_r^2 \Sigma_{yy}) = 0$$

respectively

$$(\Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx} - \varrho_r^2 \Sigma_{xx}) b_r = 0$$

$$\det (\Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx} - \varrho_r^2 \Sigma_{xx}) = 0$$

- Note that Theorem 6.12 concerns ϱ , and Theorem 6.13 concerns ϱ^2 . Finding the solutions through E_1, E_2 thus means that the canonical correlation ϱ_r is obtained as the square root of the eigenvalue ϱ_r^2 .

Question 1.6

Question 1.6

Using a linear combination of the temperature variables, the maximum fraction of variance that can be explained of a linear combination of the sound variables is:

Choose one answer

- ☐ Don't know
- ☐ 0.450
- ☐ 0.302
- ☐ 0.961
- ☐ 0.010
- ☐ 0.403

- We find the E_1 matrix and derive the largest eigenvalue: With $Y = \text{sound}$ and $X = \text{temperature}$, we have that

$$E_1 = \Sigma_{yy}^{-1} \Sigma_{yx} \Sigma_{xx}^{-1} \Sigma_{xy}$$

Question 1.6

Question 1.6

Using a linear combination of the temperature variables, the maximum fraction of variance that can be explained of a linear combination of the sound variables is:

Choose one answer

- ☐ Don't know
- ☐ 0.450
- ☐ 0.302
- ☐ 0.961
- ☐ 0.010
- ☐ 0.403

Coding:

```
temperature<-occupancy[,1:4]
sound<-occupancy[,9:12]
```

```
R11<-cor(sound)
R22<-cor(temperature)
R12<-cor(sound,temperature)
R21<-t(R12)
```

```
E1<-solve(R11)%*%R12*solve(R22)%*%R21
eigen(E1)$values
```

```
[1] 0.3020412322 0.0103145117 0.0067477629 0.0005432065
```

- 0.302, option 3.

Problem 2

Problem 2 - Room occupancy, discrimination

We still consider the data from Problem 1, but now include the number of people present in the room: *RoomCount*

We now want to see if we can discriminate between the number of people present based on the Sound and Temp variables.

The occupancy of the room varies between 0 and 3, and we thus have four classes.

Question 2.1

Question 2.1

Assuming equal priors and losses. The number of resubstitution misclassifications between class 0 and class 1 and vice versa - when going from Quadratic Discriminant Analysis to Linear Discriminant Analysis - increases by:

Choose one answer

- ☐ Don't know
- ☐ 315
- ☐ 673
- ☐ 299
- ☐ 412
- ☐ 29

Linear and quadratic discriminant analysis – `lda` and `qda` functions from the MASS library:

```
library(MASS)
my.lda<-lda(as.factor(RoomCount) ~ S1_Sound+S2_Sound+S3_Sound+S4_Sound+
          S1_Temp+S2_Temp+S3_Temp+S4_Temp ,
          prior=c(1/4,1/4,1/4,1/4), data=occupancy)

my.qda<-qda(as.factor(RoomCount) ~ S1_Sound+S2_Sound+S3_Sound+S4_Sound+
          S1_Temp+S2_Temp+S3_Temp+S4_Temp ,
          prior=c(1/4,1/4,1/4,1/4), data=occupancy)
```

Question 2.1

Question 2.1

Assuming equal priors and losses. The number of resubstitution misclassifications between class 0 and class 1 and vice versa - when going from Quadratic Discriminant Analysis to Linear Discriminant Analysis - increases by:

Choose one answer

- ☐ Don't know
- ☐ 315
- ☐ 673
- ☐ 299
- ☐ 412
- ☐ 29

Tabling discrepancies between predictors and truth, lda:

```
> lda.classification<-  
table(occupancy$RoomCount,predict(my.lda)[[1]],dnn=c("Nature","Classification"))  
> lda.classification
```

		Classification			
Nature		0	1	2	3
	0	7710	412	4	102
	1	76	374	9	0
	2	8	157	398	185
	3	5	25	191	473

Misclassifications between 0 and 1 and vice versa: $412+76=488$.

Question 2.1

Question 2.1

Assuming equal priors and losses. The number of resubstitution misclassifications between class 0 and class 1 and vice versa - when going from Quadratic Discriminant Analysis to Linear Discriminant Analysis - increases by:

Choose one answer

- ☐ Don't know
- ☐ 315
- ☐ 673
- ☐ 299
- ☐ 412
- ☐ 29

Tabling discrepancies between predictors and truth, qda:

```
> qda.classification<-table(occupancy$RoomCount,predict(my.qda)[[1]],dnn=c("Nature","Classification"))
> qda.classification
```

	Classification			
Nature	0	1	2	3
0	8047	152	13	16
1	37	410	12	0
2	22	101	486	139
3	45	10	157	482

Misclassifications between 0 and 1 and vice versa: $152+37=189$.

- Increased misclassifications when moving from qda to lda: $488-189=299$, option 4.

Question 2.2

Question 2.2

Assuming equal priors and losses. The overall resubstitution accuracy of Quadratic Discriminant Analysis and Linear Discriminant Analysis are: [QDA acc., LDA acc.]

Choose one answer

- ☐ [0.97, 0.93]
- ☐ [0.80, 0.74]
- ☐ [0.69, 0.68]
- ☐ [0.89, 0.81]
- ☐ [0.93, 0.88]
- ☐ Don't know

Measure	Formula	Probabilistic interpretation
Global accuracy (<i>ACC</i>)	$\frac{1}{N_{..}} \sum_{i=1}^k N_{ii}$	Probability of correct classification
Global error rate, misclassification rate (<i>MIS</i>)	$1 - \frac{1}{N_{..}} \sum_{i=1}^k N_{ii} = \frac{1}{N_{..}} \sum_{i \neq j} N_{ij}$	Probability of misclassification
Class accuracy	$\frac{N_{ii}}{N_{i.}}$	Conditional probability of being classified as π_i given true class is π_i .
Class error rate, misclassification rate	$1 - \frac{N_{ii}}{N_{i.}} = \frac{N_{i.} - N_{ii}}{N_{i.}}$	Conditional probability of not being classified as π_i given true class is π_i

Table 5.3 – Error rates estimated from the confusion matrix

- Construct the confusion matrix as in 2.1

Question 2.2

Question 2.2

Assuming equal priors and losses. The overall resubstitution accuracy of Quadratic Discriminant Analysis and Linear Discriminant Analysis are: [QDA acc., LDA acc.]

Choose one answer

☐ [0.97, 0.93]

☐ [0.80, 0.74]

☐ [0.69, 0.68]

☐ [0.89, 0.81] > # qda accuracy:

☐ [0.93, 0.88] > sum(diag(qda.classification)) /
+ sum(qda.classification)
[1] 0.9304966

> # lda accuracy:

> sum(diag(lda.classification)) /
+ sum(lda.classification)
[1] 0.8840952

```
> lda.classification
```

	Classification			
Nature	0	1	2	3
0	7710	412	4	102
1	76	374	9	0
2	8	157	398	185
3	5	25	191	473

```
> qda.classification
```

	Classification			
Nature	0	1	2	3
0	8047	152	13	16
1	37	410	12	0
2	22	101	486	139
3	45	10	157	482

- [0.93, 0.88], option 5.

Question 2.3

Question 2.3

Assuming equal priors and losses. Using LDA, we will now investigate whether the Sound variables - S1_Sound, ... , S4_Sound - provide no additional information to the discrimination between all four groups. The usual test statistic follows this distribution:

Choose one answer

- ☐ F(4, 10120)
- ☐ F(8, 10121)
- ☐ U(8, 3, 10121)
- ☐ U(4, 4, 10129)
- ☐ $\chi^2(10121)$
- ☐ Don't know

- Test for additional information:

Theorem 5.19

||| Theorem 5.19

The hypothesis that the last $p - q$ variables provide no additional information to the discrimination between the populations $\pi_1, \dots, \pi_k$ given that we are using the first q variables may be tested using the test statistic

$$\Lambda_{2|1} = \frac{|W_{2|1}|}{|T_{2|1}|} = \frac{|T_{11}|}{|T|} \times \frac{|W|}{|W_{11}|} = \frac{\Lambda_p}{\Lambda_q}$$

where Λ_p and Λ_q are values the test statistic Wilks' Lambda for the case with all p variables and for the case using the first q variables. Under the hypothesis the statistic follows a $U(p, k - 1, N - k - q)$ distribution.

Question 2.3

Question 2.3

Assuming equal priors and losses. Using LDA, we will now investigate whether the Sound variables - S1_Sound, ... , S4_Sound - provide no additional information to the discrimination between all four groups. The usual test statistic follows this distribution:

Choose one answer

- ☐ F(4, 10120)
- ☐ F(8, 10121)
- ☐ U(8, 3, 10121)
- ☐ U(4, 4, 10129)
- ☐ $\chi^2(10121)$
- ☐ Don't know

- N=10129
- 8 variables; 4 sound and 4 temperature; p=8.
- Test for additional information of the 4 sound variables: p-q=4, ie. q=4.
- 4 populations: k=4.

Test distribution:

$$\Lambda(8,10121,3) = U(8,3,10121)$$

- Option 3.

||| Theorem 5.19

The hypothesis that the last $p - q$ variables provide no additional information to the discrimination between the populations $\pi_1, \dots, \pi_k$ given that we are using the first q variables may be tested using the test statistic

$$\Lambda_{2|1} = \frac{|W_{2|1}|}{|T_{2|1}|} = \frac{|T_{11}|}{|T|} \times \frac{|W|}{|W_{11}|} = \frac{\Lambda_p}{\Lambda_q}$$

where Λ_p and Λ_q are values the test statistic Wilks' Lambda for the case with all p variables and for the case using the first q variables. Under the hypothesis the statistic follows a $U(p, k - 1, N - k - q)$ distribution.

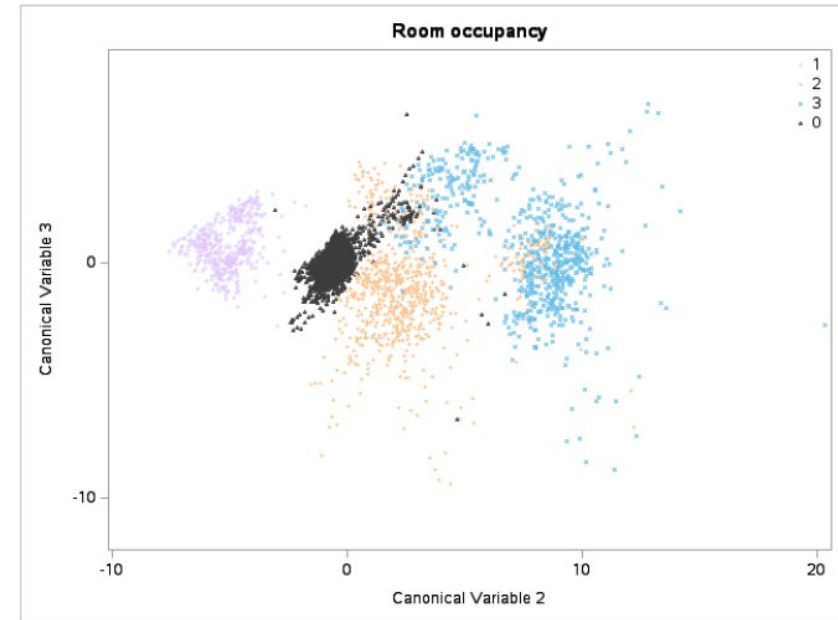
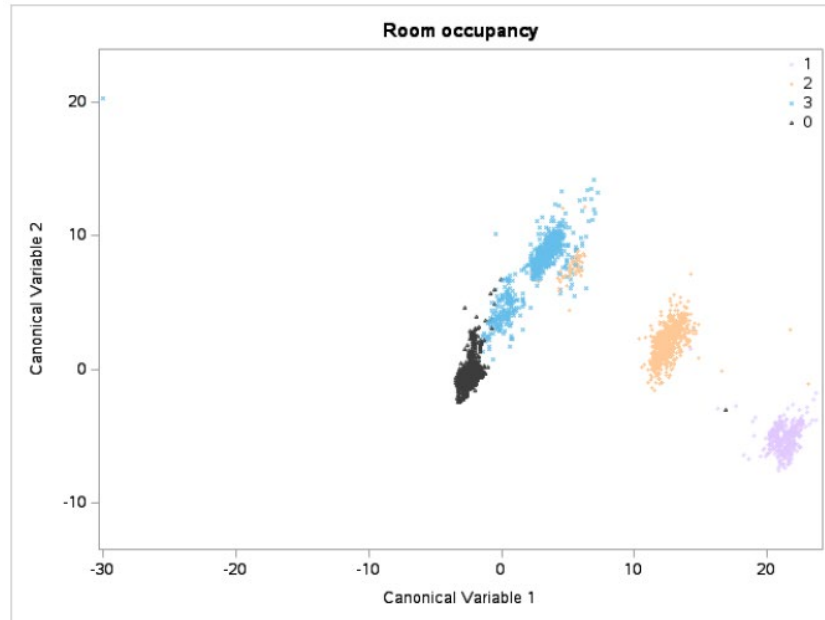
Question 2.4

Question 2.4

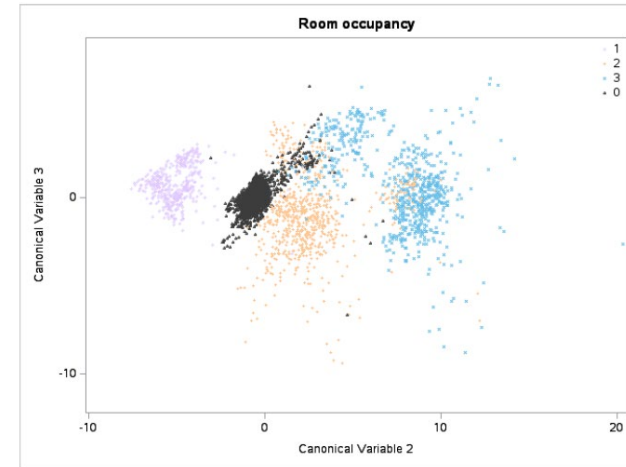
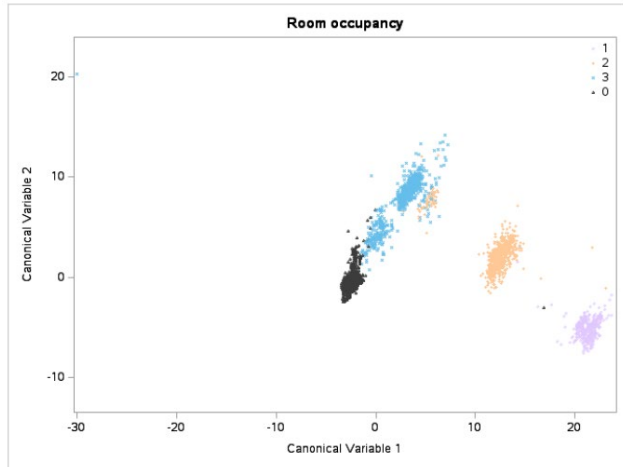
Note: In this problem, you will be provided with the necessary outputs to answer the question.

We will now consider Canonical discrimination using all variables:

S1_Temp S2_Temp S3_Temp S4_Temp S1_Light S2_Light S3_Light S4_Light S1_Sound S2_Sound
S3_Sound S4_Sound S5_CO2 S5_CO2_Slope S6_PIR S7_PIR



Question 2.4



Relying solely on the plots included in this question we can conclude:

Choose one answer

- ☐ The data are not perfectly separable with a linear classifier
- ☐ A perfectly separating hyperplane exists
- ☐ The data is perfectly separable using Quadratic Discriminant analysis.
- ☐ Don't know
- ☐ The data can be perfectly separated using the MiniMax principle
- ☐ Nothing, we need additional canonical variables shown before we can make conclusions

Option 6

NO. We haven't applied Ida.

NO. Hyperplanes separates data into two groups – we need four. And no plane even separates groups in subgroups .

NO. We haven't performed a qda

NO. We have not applied a minimax principle.

YES. While data are not separated from the pictures, we do not know if further canonical variables will separate them. 16 canonical variables...

Problem 3

Problem 3 - Room Occupancy, Factor Analysis

We will continue our analysis of the Room occupancy dataset, now considering only the variables:

S3_Light S4_Light S5_CO2 S5_CO2_Slope S6_PIR S7_PIR

We will perform a factor analysis with VARIMAX rotation. The factors is estimated using the principal factor solution.

NOTE: In this problem you will be provided with the necessary outputs to answer each of the questions. The outputs will appear in the individual questions.

Question 3.1

Question 3.1

How many factors must we include to account for 80% of the variance

	Eigenvalue	Difference	Proportion	Cumulative
1	3.06500280	2.10457684	0.5108	0.5108
2	0.96042596	0.07219669	0.1601	0.6709
3	0.88822926	0.32393981	0.1480	0.8189
4	0.56428945	0.18284511	0.0940	0.9130
5	0.38144434	0.24083616	0.0636	0.9766
6	0.14060819		0.0234	1.0000

Choose one answer

- ☐ 1
- ☐ 3
- ☐ Don't know
- ☐ 2
- ☐ 5
- ☐ 4

- Cumulative proportion is ≥ 0.80 for 3 factors – option 2.

Question 3.2

Question 3.2

	Eigenvalue	Difference	Proportion	Cumulative
1	3.06500280	2.10457684	0.5108	0.5108
2	0.96042596	0.07219669	0.1601	0.6709
3	0.88822926	0.32393981	0.1480	0.8189
4	0.56428945	0.18284511	0.0940	0.9130
5	0.38144434	0.24083616	0.0636	0.9766
6	0.14060819		0.0234	1.0000

We consider how many factors to retain. Regardless of the previous question, we consider retaining four factors. The test-statistic for the last two eigenvalues of the correlation matrix being equal, against all alternatives is:

Choose one answer

- ☐ 2423.87
- ☐ 23049.36
- ☐ 5322.25
- ☐ 4540.48
- ☐ 22124.40
- ☐ Don't know

Theorem 6.8

If we are using the estimated *variance-covariance matrix* $\hat{\Sigma}$, the test statistic for testing the hypothesis above becomes

$$Z_1 = -n^* \log \frac{\det \hat{\Sigma}}{\hat{\lambda}_1 \cdots \hat{\lambda}_m \cdot \hat{\lambda}_*^{k-m}} = -n^* \log \frac{\hat{\lambda}_{m+1} \cdots \hat{\lambda}_k}{\hat{\lambda}_*^{k-m}},$$

where

$$n^* = n - m - \frac{1}{6}(2(k - m) + 1 + \frac{2}{k - m}),$$

and

$$\hat{\lambda}_* = (\text{tr } \hat{\Sigma} - \hat{\lambda}_1 - \cdots - \hat{\lambda}_m) / (k - m) = (\hat{\lambda}_{m+1} + \cdots + \hat{\lambda}_k) / (k - m).$$

The critical region using a test at level α is approximately

$$\{(x_1, \dots, x_n) | z_1 > \chi^2(\frac{1}{2}(k - m + 2)(k - m - 1))_{1-\alpha}\}.$$

If we instead are using the estimated *correlation matrix* $\hat{R}$ we get the criterion

$$Z_2 = -n \log \frac{\det \hat{R}}{\hat{\lambda}_1 \cdots \hat{\lambda}_m \cdot \hat{\lambda}_*^{k-m}} = -n \log \frac{\hat{\lambda}_{m+1} \cdots \hat{\lambda}_k}{\hat{\lambda}_*^{k-m}},$$

where

$$\hat{\lambda}_* = (k - \hat{\lambda}_1 - \cdots - \hat{\lambda}_m) / (k - m) = (\hat{\lambda}_{m+1} + \cdots + \hat{\lambda}_k) / (k - m).$$

The critical region for a test at level α becomes approximately equal to

$$\{(x_1, \dots, x_n) | z_2 > \chi^2(\frac{1}{2}(k - m + 2)(k - m - 1))_{1-\alpha}\}.$$

However, it should be noted that this approximation is far worse than the corresponding approximation for the variance-covariance matrix.

Question 3.2

	Eigenvalue	Difference	Proportion	Cumulative
1	3.06500280	2.10457684	0.5108	0.5108
2	0.96042596	0.07219669	0.1601	0.6709
3	0.88822926	0.32393981	0.1480	0.8189
4	0.56428945	0.18284511	0.0940	0.9130
5	0.38144434	0.24083616	0.0636	0.9766
6	0.14060819		0.0234	1.0000

We consider how many factors to retain. Regardless of the previous question, we consider retaining four factors. The test-statistic for the last two eigenvalues of the correlation matrix being equal, against all alternatives is:

Choose one answer

- ☐ 2423.87
☐ 23049.36
☐ 5322.25
☐ 4540.48
☐ 22124.40
☐ Don't know

- Coding:

```
> (n<-dim(occupancy)[1])
[1] 10129
```

```
> #k=number of variables
> k<-6
> (m<-k-2)
[1] 4
```

```
> lambda.5<-0.38144434
> lambda.6<-0.14060819
> lambda.star<-(lambda.5+lambda.6)/(k-m)
> # test statistic:
> -n*log(lambda.5*lambda.6/lambda.star^(k-m))
[1] 2423.869
```

- Option 1.

Question 3.2

Theorem 6.8

If we are using the estimated *variance-covariance matrix* $\hat{\Sigma}$, the test statistic for testing the hypothesis above becomes

$$Z_1 = -n^* \log \frac{\det \hat{\Sigma}}{\hat{\lambda}_1 \cdots \hat{\lambda}_m \cdot \hat{\lambda}_*^{k-m}} = -n^* \log \frac{\hat{\lambda}_{m+1} \cdots \hat{\lambda}_k}{\hat{\lambda}_*^{k-m}},$$

where

$$n^* = n - m - \frac{1}{6}(2(k-m) + 1 + \frac{2}{k-m}),$$

and

$$\hat{\lambda}_* = (\text{tr} \hat{\Sigma} - \hat{\lambda}_1 - \cdots - \hat{\lambda}_m) / (k - m) = (\hat{\lambda}_{m+1} + \cdots + \hat{\lambda}_k) / (k - m).$$

The critical region using a test at level α is approximately

$$\{(x_1, \dots, x_n) | z_1 > \chi^2(\frac{1}{2}(k-m+2)(k-m-1))_{1-\alpha}\}.$$

If we instead are using the estimated *correlation matrix* $\hat{R}$ we get the criterion

$$Z_2 = -n \log \frac{\det \hat{R}}{\hat{\lambda}_1 \cdots \hat{\lambda}_m \cdot \hat{\lambda}_*^{k-m}} = -n \log \frac{\hat{\lambda}_{m+1} \cdots \hat{\lambda}_k}{\hat{\lambda}_*^{k-m}},$$

where

$$\hat{\lambda}_* = (k - \hat{\lambda}_1 - \cdots - \hat{\lambda}_m) / (k - m) = (\hat{\lambda}_{m+1} + \cdots + \hat{\lambda}_k) / (k - m).$$

The critical region for a test at level α becomes approximately equal to

$$\{(x_1, \dots, x_n) | z_2 > \chi^2(\frac{1}{2}(k-m+2)(k-m-1))_{1-\alpha}\}.$$

However, it should be noted that this approximation is far worse than the corresponding approximation for the variance-covariance matrix.

Question 3.3

Question 3.3

Initial Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.89348	-0.04640	0.23224
S4_Light	0.55951	0.37313	0.71287
S5_CO2	0.66549	-0.66820	0.05554
S5_CO2_Slope	0.57004	0.60222	-0.41397
S6_PIR	0.75198	0.01530	-0.17995
S7_PIR	0.78763	-0.09830	-0.34537

We consider a factor analysis as described in the problem.
The initial factor 2 and factor 3 fits the following description:

Choose one answer

- ☐ Factor 2 is an average of S3_light, S6_PIR and S7_PIR. Factor3 is an average of S3_Light and S4_Light.
- ☐ Don't know
- ☐ Factor 2 is mainly a contrast of S5_CO2 against S4_Light and S5_CO2_Slope. Factor 3 is mainly an average of all the variables
- ☐ Factor 2 is a sum of S4_light and S5_CO2_Slope. Factor 3 is a contrast of S5_CO2 against S4_Light and S5_CO2_Slope.
- ☐ Factor 2 is mainly a contrast of S5_CO2 against S4_Light and S5_CO2_Slope. Factor 3 is mainly a contrast between S3_Light and S4_Light against S6_PIR, S7_PIR, and S5_CO2_Slope
- ☐ Factor 2 is mainly an average and Factor 3 is mainly S5_CO2_Slope

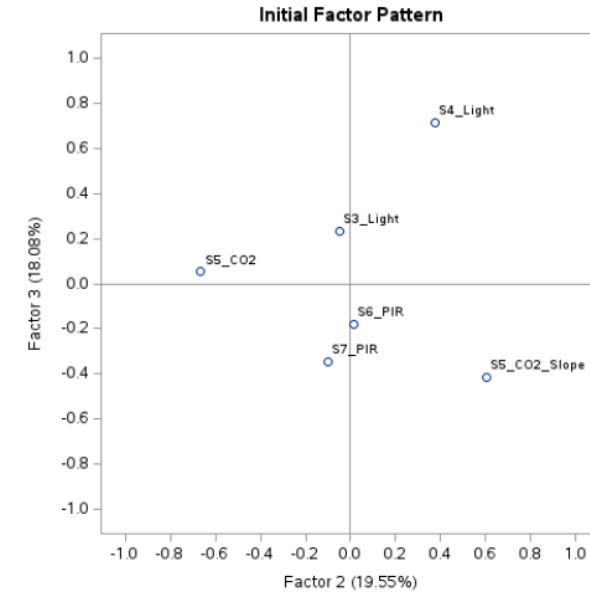
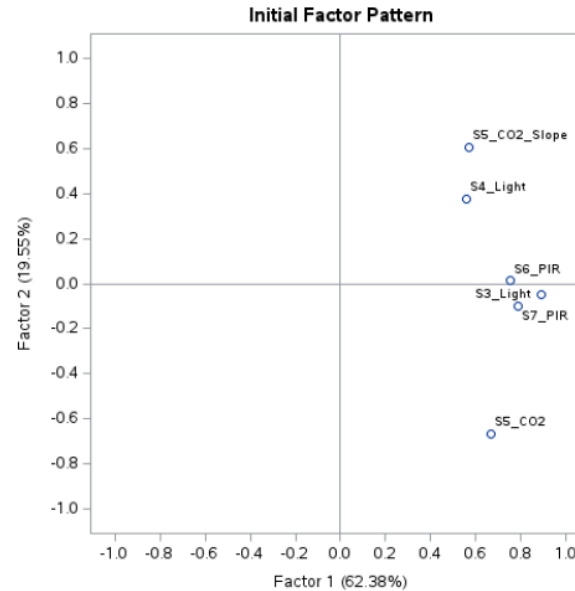
NO

NO

NO

Yes

NO



Graphical observations:

Option 5.

Question 3.4

Question 3.4

Rotated Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.64295	0.33783	0.57173
S4_Light	0.08579	0.11449	0.96953
S5_CO2	0.93899	-0.05035	0.09059
S5_CO2_Slope	-0.01442	0.91454	0.14963
S6_PIR	0.52105	0.53604	0.19813
S7_PIR	0.63560	0.58646	0.03727

Final Commuality Estimates

S3_Light	S4_Light	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR
0.85438728	0.96045827	0.89244227	0.85897641	0.59808453	0.74930926

We now consider the rotated factors. Factor2 and Factor3 describes the following fraction of the variance of S4_Light

Choose one answer

- ☐ 0.953
- ☐ 0.854
- ☐ 0.960/3
- ☐ $(0.11449+0.96953)/3$
- ☐ Don't know
- ☐ 0.960

```
> 0.11449^2+0.96953^2
[1] 0.9530964
```

- $V(X)=I$ assumed; option 1.

Question 3.5

Question 3.5

Initial Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.89348	-0.04640	0.23224
S4_Light	0.55951	0.37313	0.71287
S5_CO2	0.66549	-0.66820	0.05554
S5_CO2_Slope	0.57004	0.60222	-0.41397
S6_PIR	0.75198	0.01530	-0.17995
S7_PIR	0.78763	-0.09830	-0.34537

The correlation matrix for the original variables is

	S3_Light	S4_Light	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR
S3_Light	1	0.57948	0.65083	0.44771	0.50184	0.57781
S4_Light	0.57948	1	0.14861	0.21272	0.32454	0.22020
S5_CO2	0.65083	0.14861	1	0.06922	0.39526	0.47344
S5_CO2_Slope	0.44771	0.21272	0.06922	1	0.36837	0.42535
S6_PIR	0.50184	0.32454	0.39526	0.36837	1	0.57113
S7_PIR	0.57781	0.22020	0.47344	0.42535	0.57113	1

Note: If it is helpful for your calculations, you are welcome to use statistical software to recreate the correlation matrix.

We now consider the initial factors. The covariance between initial factor 1 and initial factor 2 is:

Choose one answer

- ☐ 0
☐ 5.02
☐ 3.30
☐ 6.53
☐ 9.39
☐ Don't know

Option 1 – 0. Initial factors are always assumed to be uncorrelated.

Chapter 6 ||| 6.3 FACTOR ANALYSIS

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Here we note that F_i and G_i change value when the observations X_i change value. We can aggregate the above models into

$$\begin{bmatrix} X_{11} \cdots X_{n1} \\ \vdots \\ X_{k1} \cdots X_{nk} \end{bmatrix} = \begin{bmatrix} a_{11} \cdots a_{1m} \\ \vdots \\ a_{k1} \cdots a_{km} \end{bmatrix} \begin{bmatrix} F_{11} \cdots F_{n1} \\ \vdots \\ F_{1m} \cdots F_{nm} \end{bmatrix} + \begin{bmatrix} G_{11} \cdots G_{n1} \\ \vdots \\ G_{1k} \cdots G_{nk} \end{bmatrix}.$$

It is assumed that F and G are uncorrelated and that

$$D(F) = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 1 \end{pmatrix} = I = I_m,$$

Initial Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.89348	-0.04640	0.23224
S4_Light	0.55951	0.37313	0.71287
S5_CO2	0.66549	-0.66820	0.05554
S5_CO2_Slope	0.57004	0.60222	-0.41397
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Rotated Factor Pattern

	Factor1	Factor2	Factor3
S3_Light	0.64295	0.33783	0.57173
S4_Light	0.08579	0.11449	0.96953
S5_CO2	0.93899	-0.05035	0.09059
S5_CO2_Slope	-0.01442	0.91454	0.14963
S6_PIR	0.52105	0.53604	0.19813
S7_PIR	0.63560	0.58646	0.03727

The correlation matrix for the original variables is

	S3_Light	S4_Light	S5_CO2	S5_CO2_Slope	S6_PIR	S7_PIR
S3_Light	1	0.57948	0.65083	0.44771	0.50184	0.57781
S4_Light	0.57948	1	0.14861	0.21272	0.32454	0.22020
S5_CO2	0.65083	0.14861	1	0.06922	0.39526	0.47344
S5_CO2_Slope	0.44771	0.21272	0.06922	1	0.36837	0.42535
S6_PIR	0.50184	0.32454	0.39526	0.36837	1	0.57113
S7_PIR	0.57781	0.22020	0.47344	0.42535	0.57113	1

Note: If it is helpful for your calculations, you are welcome to use statistical software to recreate the correlation matrix.

We now consider both the initial and the rotated factors. The covariance between initial factor 1 and rotated factor 1 is:

Choose one answer

- ☐ 3.30
- ☐ 5.02
- ☐ 6.53
- ☐ 0
- ☐ 9.39
- ☐ Don't know

Question 3.6

Factors solutions are displayed as functions of X , ie. $F = C_F X$, $RF = C_{RF} X$. Thus

$$\text{Cov}(F, RF) = \text{Cov}(C_F X, C_{RF} X) = C_F V(X) C_{RF}^T$$

Coding:

```
> V1.1<-c(0.89348,0.55951,0.66549,0.57004,0.75198,0.78763)
```

```
> V2.1<-c(0.64295,0.08579,0.93899,-0.01442,0.52105,0.63560)
```

```
> My.V<-matrix(c(1,0.57948,0.65083,0.44771,0.50184,0.57781,
+               0.57948,1,0.14861,0.21272,0.32454,0.22020,
+               0.65083,0.14861,1,0.06922,0.39526,0.47344,
+               0.44771,0.21272,0.06922,1,0.36837,0.42535,
+               0.50184,0.32454,0.39526,0.36837,1,0.57113,
+               0.57781,0.22020,0.47344,0.42535,0.57113,1),byrow=T,nrow=6)
```

```
> V1.1*%My.V*%t(t(V2.1))
```

```
      [,1]
```

```
[1,] 6.533266
```

- Option 3 – 6.53

Problem 4

PROBLEM 4 - Toxicity

We consider the QSAR fish toxicity study from here: <https://archive.ics.uci.edu/dataset/504/qsar+fish+toxicity>
The data we will use is available here in R (<https://designer.mcq.eksamen.dtu.dk/api/images/55d44bce-fd41-406f-8af0-fbb939c13972>) and in SAS (<https://designer.mcq.eksamen.dtu.dk/api/images/06e393b2-da5e-4faf-a895-65fd5aacae89>) .

The data set contains values for 6 attributes (molecular descriptors) of 908 chemicals used to predict quantitative acute aquatic toxicity towards the fish *Pimephales promelas* (fathead minnow). The outcome variable is LC50, the contraction at which 50 % of the fish die.

The original variable names are not important, and has been replaced as follows:
X1 = CIC0, X2 = SM1Dz, X3 = GATS1i, X4 = NdsCH, X5 = NdssC, X6 = MLOGP.

We will consider the model:

$$LC50 = \alpha + X1 + X2 + X3 + X4 + X5 + X6 + \epsilon$$

where α is an intercept and ϵ is the residuals

Problem 4

PROBLEM 4 - Toxicity

We consider the QSAR fish toxicity study from here: <https://archive.ics.uci.edu/dataset/504/qsar+fish+toxicity>
The data we will use is available here in R (<https://designer.mcq.eksamen.dtu.dk/api/images/55d44bce-fd41-406f-8af0-fbb939c13972>) and in SAS (<https://designer.mcq.eksamen.dtu.dk/api/images/06e393b2-da5e-4faf-a895-65fd5aacae89>).

The data set contains values for 6 attributes (molecular descriptors) of 908 chemicals used to predict quantitative acute aquatic toxicity towards the fish *Pimephales promelas* (fathead minnow). The outcome variable is LC50, the contraction at which 50 % of the fish die.

The original variable names are not important, and has been replaced as follows:
X1 = CIC0, X2 = SM1Dz, X3 = GATS1i, X4 = NdsCH, X5 = NdssC, X6 = MLOGP.

We will consider the model:

$$LC50 = \alpha + X1 + X2 + X3 + X4 + X5 + X6 + \epsilon$$

where α is an intercept and ϵ is the residuals

Initial coding:

```
> toxicity<-read.csv2("qsar_fish_toxicity3.csv")
> names(toxicity)
[1] "CIC0"      "SM1_Dz.Z." "GATS1i"     "NdsCH"      "NdssC"      "MLOGP"
[7] "LC50"
> names(toxicity)[1:6]<-paste("X",1:6,sep="")
> analysis<-lm(LC50~X1+X2+X3+X4+X5+X6,data=toxicity)
> summary(analysis)
```

Call:

```
lm(formula = LC50 ~ X1 + X2 + X3 + X4 + X5 + X6, data = toxicity)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.4921	-0.5287	-0.0712	0.4861	5.6876

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.17456	0.18122	12.000	< 2e-16	***
X1	0.38563	0.06089	6.333	3.79e-10	***
X2	1.25562	0.08702	14.430	< 2e-16	***
X3	-0.74641	0.10135	-7.365	4.00e-13	***
X4	0.41355	0.05410	7.644	5.41e-14	***
X5	0.06433	0.04064	1.583	0.114	
X6	0.39005	0.03376	11.555	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9497 on 901 degrees of freedom
Multiple R-squared: 0.5772, Adjusted R-squared: 0.5743
F-statistic: 205 on 6 and 901 DF, p-value: < 2.2e-16

Question 4.1

Question 4.1

The R^2 of the model is:

Choose one answer

☐ 0.9019

☐ 0.5772

☐ Don't know

☐ 0.3901

☐ 0.4135

☐ 0.5743

```
> summary(analysis)
```

```
Call:
```

```
lm(formula = LC50 ~ X1 + X2 + X3 + X4 + X5 + X6, data = toxicity)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-4.4921 -0.5287 -0.0712  0.4861  5.6876
```

```
Coefficients:
```

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.17456    0.18122   12.000 < 2e-16 ***
X1           0.38563    0.06089    6.333 3.79e-10 ***
X2           1.25562    0.08702   14.430 < 2e-16 ***
X3          -0.74641    0.10135   -7.365 4.00e-13 ***
X4           0.41355    0.05410    7.644 5.41e-14 ***
X5           0.06433    0.04064    1.583  0.114
X6           0.39005    0.03376   11.555 < 2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9497 on 901 degrees of freedom
```

```
Multiple R-squared:  0.5772,    Adjusted R-squared:  0.5743
```

```
F-statistic:    205 on 6 and 901 DF,  p-value: < 2.2e-16
```

- 0.5772 – option 2.

Question 4.2

Question 4.2

The number of influential observations in the model, i.e., observations with both an absolute RSTUDENT larger than 2 and a leverage larger than 0.015 are:

Choose one answer

- ☐ 23
- ☐ 8
- ☐ 4
- ☐ Don't know
- ☐ 17
- ☐ 22

```
?rstudent
```

```
?leverage
```

We learn that we can use the `hat` function.

```
> X<-as.matrix(toxicity[,1:6])
> sum( (abs(rstudent(analysis))>=2) * (hat(X,intercept=T)>0.015) )
[1] 8
```

- Option 2

Page 205-206:

RSTUDENT & STUDENT RESIDUAL

RSTUDENT is a so-called “studentised” residual, i.e.

$$RSTUDENT_i = \frac{r_i}{\hat{\sigma}(i)\sqrt{1-h_{ii}}},$$

where $\hat{\sigma}(i)^2$ is the estimate of variance corresponding to deletion of the i 'th observation.

Leverage

The quantity h_{ii} introduced earlier is called the *leverage*. It is

$$h_{ii} = \mathbf{x}_i(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_i^T$$

Question 4.3

Question 4.3

We now consider the individual parameter estimates. Using DFBETAS, the following variables have observations with an absolute DFBETAS larger than 0.4

Choose one answer

- ☐ X2, X4, X5
- ☐ X1, X3, X6
- ☐ X2, X4, X6
- ☐ X4, X5
- ☐ X1, X4, X6
- ☐ Don't know

Learned from ?rstudent about dfbetas:

```
> temp<-dfbetas(analysis)
> temp<-abs(temp)
> apply(temp,2,max)
```

(Intercept)	X1	X2	X3	X4	X5	X6
0.3405429	0.2708175	0.5152537	0.3910852	0.5301052	0.3209490	0.5658828

- X_2, X_4, X_6 - option 3

Question 4.4

Question 4.4

We now consider multicollinearity. The following is true:

Choose one answer

- ☐ The TOL for X1 is less than 0.1 which indicates collinearity
- ☐ The TOL for X1 is larger than 10 which indicates collinearity
- ☐ The VIF for X6 is less than 0.1 which indicates collinearity
- ☐ There is no indication of multicollinearity
- ☐ The VIF for X6 is larger than 10 which indicates collinearity
- ☐ Don't know

Definition 3.15

We define the *tolerance (TOL)* and the *variance inflation (VIF)* as

$$\begin{aligned} \text{TOL}_i &= 1 - R^2(x_i | \text{all other } x\text{-variables}) \\ \text{VIF}_i &= \frac{1}{\text{TOL}_i} \end{aligned}$$

As a rule of thumb, $\text{TOL} < 0.1$ or equivalently $\text{VIF} > 10$ indicates a multicollinearity problem.

Question 4.4

Question 4.4

We now consider multicollinearity. The following is true:

Choose one answer

- ☐ The TOL for X_1 is less than 0.1 which indicates collinearity
- ☐ The TOL for X_1 is larger than 10 which indicates collinearity
- ☐ The VIF for X_6 is less than 0.1 which indicates collinearity
- ☐ There is no indication of multicollinearity
- ☐ The VIF for X_6 is larger than 10 which indicates collinearity
- ☐ Don't know

Tolerance of X_1 :

```
> 1-summary(lm(X1~X2+X3+X4+X5+X6,data=toxicity))$r.squared  
[1] 0.4691623
```

Larger than 0.1

Tolerance of X_6 :

```
> 1-summary(lm(X6~X1+X2+X3+X4+X5,data=toxicity))$r.squared  
[1] 0.4248685
```

Larger than 0.1

Tolerance of $X_2 - X_5$:

```
> 1-summary(lm(X2~X1+X3+X4+X5+X6,data=toxicity))$r.squared  
[1] 0.7154447  
> 1-summary(lm(X3~X1+X2+X4+X5+X6,data=toxicity))$r.squared  
[1] 0.6227454  
> 1-summary(lm(X4~X1+X2+X3+X5+X6,data=toxicity))$r.squared  
[1] 0.9271029  
> 1-summary(lm(X5~X1+X2+X3+X4+X6,data=toxicity))$r.squared  
[1] 0.8117598
```

- Option 4

Question 4.5

Question 4.5

When performing a forward selection, the first variable to include is:

Choose one answer

- ☐ X5
- ☐ X1
- ☐ X3
- ☐ Don't know
- ☐ X6
- ☐ X4

- Forward selection is in general no go – will not disclose functions in R 😊.

Question 4.5

When performing a forward selection, the first variable to include is:

Choose one answer

- ☐ X5
- ☐ X1
- ☐ X3
- ☐ Don't know
- ☐ X6
- ☐ X4

```
> drop1(lm(toxicity$LC50~toxicity[,1]),test="F")
Single term deletions

Model:
toxicity$LC50 ~ toxicity[, 1]
              Df Sum of Sq  RSS    AIC F value    Pr(>F)
<none>                    1758.3 604.05
toxicity[, 1]  1      163.71 1922.0 682.88  84.358 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> drop1(lm(toxicity$LC50~toxicity[,2]),test="F")
Single term deletions

Model:
toxicity$LC50 ~ toxicity[, 2]
              Df Sum of Sq  RSS    AIC F value    Pr(>F)
<none>                    1597.5 516.97
toxicity[, 2]  1      324.5 1922.0 682.88  184.03 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> drop1(lm(toxicity$LC50~toxicity[,3]),test="F")
Single term deletions

Model:
toxicity$LC50 ~ toxicity[, 3]
              Df Sum of Sq  RSS    AIC F value    Pr(>F)
<none>                    1617.6 528.32
toxicity[, 3]  1      304.4 1922.0 682.88  170.49 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 4.5

```
> drop1(lm(toxicity$LC50~toxicity[,4]),test="F")
Single term deletions
```

```
Model:
toxicity$LC50 ~ toxicity[, 4]
              Df Sum of Sq  RSS    AIC F value    Pr(>F)
<none>                    1865.1 657.61
toxicity[, 4]  1      56.863 1922.0 682.88  27.622 1.84e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> drop1(lm(toxicity$LC50~toxicity[,5]),test="F")
Single term deletions
```

```
Model:
toxicity$LC50 ~ toxicity[, 5]
              Df Sum of Sq  RSS    AIC F value    Pr(>F)
<none>                    1864.9 657.49
toxicity[, 5]  1      57.118 1922.0 682.88  27.749 1.726e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> drop1(lm(toxicity$LC50~toxicity[,6]),test="F")
Single term deletions
```

```
Model:
toxicity$LC50 ~ toxicity[, 6]
              Df Sum of Sq  RSS    AIC F value    Pr(>F)
<none>                    1105.8 182.94
toxicity[, 6]  1      816.2 1922.0 682.88  668.74 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Largest F-value: X_6 .
- Option 5

Question 4.6

Question 4.6

We want to estimate a 95%-confidence interval for the expected value of observation 1. Given new additional information we have been informed that the variance can be assumed to be $V(U) = \sigma^2 \cdot 0.01$. The 95%-confidence interval is then:

Choose one answer

- ☐ [3.74, 3.96]
- ☐ [3.77, 3.93]
- ☐ Don't know
- ☐ [3.58, 3.96]
- ☐ [3.70, 4.01]
- ☐ [3.66, 4.04]

- Expected value is found with the `predict` function:

```
> my.mu<-predict(analysis)[1]  
> my.sd<-sqrt(summary(analysis)$sigma^2*0.01)
```

```
> my.mu-1.96*my.sd
```

1

```
3.666557
```

```
> my.mu+1.96*my.sd
```

1

```
4.03885
```

- [3.67;4.04], (nearly) option 6.

Question 4.7

Question 4.7

Regardless of the previous question we now consider observation 6.

We want a prediction interval for a new observation and due to degradation of sensors the variance has increased and the new observation Y_q follows $Y_q \sim N(E(Y), 2 \cdot \sigma^2)$. The 95% prediction interval is

Choose one answer

- ☐ [-0.16, 5.12]
- ☐ Don't know
- ☐ [0.27, 4.70]
- ☐ [2.32, 2.64]
- ☐ [0.61, 4.35]
- ☐ [-0.82, 4.46]

- Completely similar to 4.6:

```
> my.mu<-predict (analysis) [6]
> my.sd<-sqrt (2*summary (analysis) $sigma^2)

> my.mu-1.96*my.sd
      6
-0.149416
> my.mu+1.96*my.sd
      6
5.115603
```

- [-0.15;5.12], (nearly) option 1.

Question 4.8

Question 4.8

We now propose a new model where we remove X1 and X6. When we test the original model against the new proposed model the resulting F-value is:

Choose one answer

- ☐ 247.29
- ☐ Don't know
- ☐ 160.01
- ☐ 118.94
- ☐ 204.97
- ☐ 0.3568

- We use the `anova` function to compare the models:

```
> analysis2<-lm(LC50~X2+X3+X4+X5,data=toxicity)
```

```
> anova(analysis2,analysis)
```

```
Analysis of Variance Table
```

```
Model 1: LC50 ~ X2 + X3 + X4 + X5
```

```
Model 2: LC50 ~ X1 + X2 + X3 + X4 + X5 + X6
```

```
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     903 1258.78
2     901  812.69  2    446.1 247.29 < 2.2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- 247.29 – option 1.

Problem 5

Problem 5

We consider the following model

$$\begin{bmatrix} Y_1 & Z_1 \\ Y_2 & Z_2 \\ Y_3 & Z_3 \\ Y_4 & Z_4 \\ Y_5 & Z_5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} \alpha_y & \alpha_z \\ \beta_y & \beta_z \\ \gamma_y & \gamma_z \end{bmatrix} + \begin{bmatrix} \delta_1 & \varepsilon_1 \\ \delta_2 & \varepsilon_2 \\ \delta_3 & \varepsilon_3 \\ \delta_4 & \varepsilon_4 \\ \delta_5 & \varepsilon_5 \end{bmatrix}$$

where the error terms $[\delta_i \ \varepsilon_i]$ for $i=1,\dots,5$ are independent and normally distributed $N_2(\mathbf{0}, \mathbf{\Sigma})$, and where $\mathbf{\Sigma}$ is the unknown dispersion matrix.

We have obtained the following observations

$$\begin{bmatrix} 21 & 17 \\ 16 & 12 \\ 17 & 13 \\ 19 & 16 \\ 25 & 19 \end{bmatrix}$$

and the unbiased estimate of $\mathbf{\Sigma}$ is

$$\begin{bmatrix} 1.13333333 & 0.36666667 \\ 0.36666667 & 0.13333333 \end{bmatrix}$$

- Initial coding:

```
> X<-cbind(rep(1,5), c(1,1,0,2,0),c(1,0,1,0,2))
> Y<-cbind(c(21,16,17,19,25),c(17,12,13,16,19))
> Sigma<-matrix(c(17/15,11/30,11/30,2/15),nrow=2)
```

```
> # Doesn't work:
```

```
> #Sigma<-matrix(c(1.3333333,0.36666667,0.36666667,0.13333333),nrow=2)
```

Question 5.1

Question 5.1

The maximum likelihood estimate for the parameters $[\alpha_z \quad \beta_z \quad \gamma_z]$ are:

Choose one answer

- ☐ [11.33, 3.87, 6.47]
- ☐ Don't know
- ☐ [7.7, 4.1, 5.5]
- ☐ [0.31, -0.18, -0.18]
- ☐ [2.64, -1.51, -1.51]
- ☐ [9.67, 5.51, 4.47]

||| Theorem 4.14

We consider the above mentioned situation. If the observations Y_i are normally distributed the maximum likelihood estimate of θ is given by

$$\hat{\theta} = (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \mathbf{Y}.$$

Question 5.1

Question 5.1

The maximum likelihood estimate for the parameters $[\alpha_z \ \beta_z \ \gamma_z]$ are:

Choose one answer

- ☐ [11.33, 3.87, 6.47]
- ☐ Don't know
- ☐ [7.7, 4.1, 5.5]
- ☐ [0.31, -0.18, -0.18]
- ☐ [2.64, -1.51, -1.51]
- ☐ [9.67, 5.51, 4.47]

```
> (thetahat<-solve(t(X)%*%X)%*%t(X)%*%Y)
      [,1]      [,2]
[1,] 11.333333  7.666667
[2,]  3.866667  4.133333
[3,]  6.466667  5.533333
```

- (7.7,4.1,5.5) - option 3.

Question 5.2

Question 5.2

The estimated covariance between α_y and α_z is:

Choose one answer

- ☐ 0.86
- ☐ -1.51
- ☐ Don't know
- ☐ 0.37
- ☐ 0.31
- ☐ 1.13

|||| Theorem 4.19

We consider the situation from theorems 4.14 and 4.18 and we introduce the usual notations

$$\begin{aligned}\tilde{\boldsymbol{\theta}} &= \text{vc}(\boldsymbol{\theta}) = \begin{bmatrix} \boldsymbol{\theta}_{|1} \\ \vdots \\ \boldsymbol{\theta}_{|p} \end{bmatrix} \\ \hat{\tilde{\boldsymbol{\theta}}} &= \text{vc}(\hat{\boldsymbol{\theta}}) = \begin{bmatrix} \hat{\boldsymbol{\theta}}_{|1} \\ \vdots \\ \hat{\boldsymbol{\theta}}_{|p} \end{bmatrix}.\end{aligned}$$

Then we have that $\hat{\tilde{\boldsymbol{\theta}}}$ is normally distributed

$$\hat{\tilde{\boldsymbol{\theta}}} = \text{vc}(\hat{\boldsymbol{\theta}}) \sim N_{pk}(\tilde{\boldsymbol{\theta}}, \boldsymbol{\Sigma} \otimes (\mathbf{x}^T \mathbf{x})^{-1}),$$

and $n\hat{\boldsymbol{\Sigma}}^*$ is Wishart distributed

$$n\hat{\boldsymbol{\Sigma}}^* \sim W(n - k, \boldsymbol{\Sigma}).$$

Finally $\hat{\boldsymbol{\Sigma}}^*$ and $\hat{\tilde{\boldsymbol{\theta}}}$ and therefore also $\hat{\boldsymbol{\Sigma}}^*$ and $\hat{\boldsymbol{\theta}}$ are stochastically independent.

- Remember notation in the book: data as columns not rows.

Question 5.2

Question 5.2

The estimated covariance between α_y and α_z is:

Choose one answer

- ☐ 0.86
- ☐ -1.51
- ☐ Don't know
- ☐ 0.37
- ☐ 0.31
- ☐ 1.13

Note that

$$V(\hat{\theta}) = (X^T X)^{-1} \otimes \Sigma$$

$$((X^T X)^{-1} \otimes \Sigma)_{ir, js} = ((X^T X)^{-1})_{ij} \cdot \Sigma_{rs}$$

$$\text{Cov}(\hat{\alpha}_y, \hat{\alpha}_z) = \text{Cov}(\hat{\theta}_{11}, \hat{\theta}_{12}) = ((X^T X)^{-1} \otimes \Sigma)_{11, 12} = ((X^T X)^{-1})_{11} \cdot \Sigma_{12}$$

Code:

```
> solve(t(X) %*% X) [1, 1] * Sigma [1, 2]  
[1] 0.8555556
```

- 0.86 - option 1.

Question 5.3

Question 5.3

The variance of the maximum likelihood estimate of β_y is:

Choose one answer

- ☐ 0.311
- ☐ 0.124
- ☐ 1.133
- ☐ 1.058
- ☐ 2.644
- ☐ Don't know

- Similar:

$$V(\hat{\beta}) = V(\hat{\theta}_{21}) = ((X^T X)^{-1} \otimes \Sigma)_{21,21} = ((X^T X)^{-1})_{22} \cdot \Sigma_{11}$$

- Code:

```
> solve(t(X) %*% X) [2,2] * Sigma [1,1]  
[1] 1.057778
```

- 1.058 - option 4.

Question 5.4

Question 5.4

We now want to test if α_y is equal to 11 against all alternatives. The P-value is:

Choose one answer

- ☐ 0.020
- ☐ Don't know
- ☐ 0.428
- ☐ 0.857
- ☐ 0.001
- ☐ 0.987

||| Theorem 4.21

We consider the above mentioned situation including the assumption of the normality of the observations. Furthermore we consider the hypothesis

$$H_0 : \mathbf{A} \boldsymbol{\theta} \mathbf{B}^T = \mathbf{C} \quad \text{against} \quad H_1 : \mathbf{A} \boldsymbol{\theta} \mathbf{B}^T \neq \mathbf{C},$$

where $\mathbf{A}(r \times k)$, $\mathbf{B}(s \times p)$ and $\mathbf{C}(r \times s)$ are given (known) matrices. We introduce

$$\begin{aligned} \Delta &= \mathbf{A} \hat{\boldsymbol{\theta}} \mathbf{B}^T - \mathbf{C} \\ \mathbf{R} &= n \hat{\Sigma}^* = (\mathbf{Y} - \mathbf{x} \hat{\boldsymbol{\theta}})^T (\mathbf{Y} - \mathbf{x} \hat{\boldsymbol{\theta}}) = \mathbf{Y}^T \mathbf{Y} - \hat{\boldsymbol{\theta}}^T (\mathbf{x}^T \mathbf{x}) \hat{\boldsymbol{\theta}} \end{aligned}$$

and

$$\begin{aligned} \mathbf{E} &= \mathbf{B} \mathbf{R} \mathbf{B}^T \\ \mathbf{H} &= \Delta^T [\mathbf{A} (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{A}^T]^{-1} \Delta. \end{aligned}$$

The likelihood ratio test for testing H_0 against H_1 is equivalent to the test given by the critical region

$$\{y \mid \frac{\det(\mathbf{e})}{\det(\mathbf{e} + \mathbf{h})} \leq U(s, r, n - k)_\alpha\},$$

where $U(s, r, n - k)_\alpha$ is the α quantile in the null-hypothesis distribution of the test statistic (see below).

Question 5.4

Question 5.4

We now want to test if α_y is equal to 11 against all alternatives. The P-value is:

$$A = (1 \ 0 \ 0), B = (1 \ 0)$$

Choose one answer

☐ 0.020

☐ Don't know

☐ 0.428

☐ 0.857

☐ 0.001

☐ 0.987

Coding:

```
> Delta<-thetahat[1,1]-11
> R<-t(Y-X**%thetahat)%*(Y-X**%thetahat)
> (E<-R[1,1])
[1] 2.266667
> (H<-Delta*(1/solve(t(X)**X)[1,1])*Delta)
[1] 0.04761905
> (Wilks<-E/(E+H))
[1] 0.9794239
```

$n = 5$, $k = \text{rank}(X) = 3$, $r = 1$, $s = 1$ since C is a number.

Distributed as $\Lambda(1, 5 - 3, 1) = \mathcal{U}(1, 1, 5 - 3)$

$$t = \sqrt{\frac{1^2 1^2 - 4}{1^2 + 1^2 - 5}} = 1, v = \frac{1}{2}(2(5 - 3) + 1 - 1 - 1) = \frac{3}{2}$$

$$F = \frac{1-\Lambda}{\Lambda} \cdot \frac{\frac{3}{2} + 1 - \frac{1}{2}}{1} = 2 \frac{1-\Lambda}{\Lambda}$$

Coding:

```
> F<-2*(1-Wilks)/Wilks
> 1-pf(F,1,5-3)
[1] 0.8565562
```

- 0.857 - option 4

Theorem 4.21

We consider the above mentioned situation including the assumption of the normality of the observations. Furthermore we consider the hypothesis

$$H_0 : A \theta B^T = C \quad \text{against} \quad H_1 : A \theta B^T \neq C,$$

where $A(r \times k)$, $B(s \times p)$ and $C(r \times s)$ are given (known) matrices. We introduce

$$\Delta = A \hat{\theta} B^T - C$$

$$R = n \hat{\Sigma}^* = (Y - X \hat{\theta})^T (Y - X \hat{\theta}) = Y^T Y - \hat{\theta}^T (X^T X) \hat{\theta}$$

and

$$E = B R B^T$$

$$H = \Delta^T [A (X^T X)^{-1} A^T]^{-1} \Delta.$$

The likelihood ratio test for testing H_0 against H_1 is equivalent to the test given by the critical region

$$\{y \mid \frac{\det(e)}{\det(e+h)} \leq U(s, r, n-k)_\alpha\},$$

where $U(s, r, n-k)_\alpha$ is the α quantile in the null-hypothesis distribution of the test statistic (see below).

Theorem 4.22

Let U be $U(s, r, n-k)$ -distributed and let

$$t = \begin{cases} 1 & s^2 + r^2 = 5 \\ \sqrt{\frac{s^2 r^2 - 4}{s^2 + r^2 - 5}} & s^2 + r^2 \neq 5 \end{cases}$$

$$v = \frac{1}{2}(2(n-k) + r - s - 1).$$

Then

$$F = \frac{1 - U^{\frac{1}{t}}}{U^{\frac{1}{t}}} \cdot \frac{vt + 1 - \frac{1}{2}sr}{sr}$$

is approximately distributed as

$$F(sr, vt + 1 - \frac{1}{2}sr).$$

If either s or r are equal to 1 or 2, then the approximation is exact.

Question 5.5

Question 5.5

We now want to test if $[\beta_y \quad \beta_z] = [4 \quad 4]$ against all alternatives. The test statistic for this test has - under the null-hypothesis the following distribution:

Choose one answer

- ☐ U(2, 2, 2)
- ☐ U(1, 2, 2)
- ☐ U(2, 1, 3)
- ☐ U(2, 1, 2)
- ☐ U(1, 2, 3)
- ☐ Don't know

- $C = [4 \quad 4]$: $r = 1, s = 2$.

Thus, test is $\Lambda(s, n - k, r) = \mathcal{U}(s, r, n - k) = \mathcal{U}(2, 1, 2)$.

- Option 4.

Theorem 4.21

We consider the above mentioned situation including the assumption of the normality of the observations. Furthermore we consider the hypothesis

$$H_0 : \mathbf{A} \boldsymbol{\theta} \mathbf{B}^T = \mathbf{C} \quad \text{against} \quad H_1 : \mathbf{A} \boldsymbol{\theta} \mathbf{B}^T \neq \mathbf{C},$$

where $\mathbf{A}(r \times k)$, $\mathbf{B}(s \times p)$ and $\mathbf{C}(r \times s)$ are given (known) matrices. We introduce

$$\begin{aligned} \Delta &= \mathbf{A} \hat{\boldsymbol{\theta}} \mathbf{B}^T - \mathbf{C} \\ \mathbf{R} &= n \hat{\Sigma}^* = (\mathbf{Y} - \mathbf{x} \hat{\boldsymbol{\theta}})^T (\mathbf{Y} - \mathbf{x} \hat{\boldsymbol{\theta}}) = \mathbf{Y}^T \mathbf{Y} - \hat{\boldsymbol{\theta}}^T (\mathbf{x}^T \mathbf{x}) \hat{\boldsymbol{\theta}} \end{aligned}$$

and

$$\begin{aligned} \mathbf{E} &= \mathbf{B} \mathbf{R} \mathbf{B}^T \\ \mathbf{H} &= \Delta^T [\mathbf{A} (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{A}^T]^{-1} \Delta. \end{aligned}$$

The likelihood ratio test for testing H_0 against H_1 is equivalent to the test given by the critical region

$$\left\{ \mathbf{y} \mid \frac{\det(\mathbf{e})}{\det(\mathbf{e} + \mathbf{h})} \leq \mathcal{U}(s, r, n - k)_\alpha \right\},$$

where $\mathcal{U}(s, r, n - k)_\alpha$ is the α quantile in the null-hypothesis distribution of the test statistic (see below).

Question 5.6

Question 5.6

We now want to test if $[\alpha_y \quad \beta_y \quad \gamma_y] = [11 \quad 4 \quad 6]$ against all alternatives. Wilks Lambda for this test is:

Choose one answer

- ☐ 0.453
- ☐ 0.222
- ☐ Don't know
- ☐ 0.004
- ☐ 0.586
- ☐ 0.001

- Similar to 5.4.

||| Theorem 4.21

We consider the above mentioned situation including the assumption of the normality of the observations. Furthermore we consider the hypothesis

$$H_0 : \mathbf{A} \boldsymbol{\theta} \mathbf{B}^T = \mathbf{C} \quad \text{against} \quad H_1 : \mathbf{A} \boldsymbol{\theta} \mathbf{B}^T \neq \mathbf{C},$$

where $\mathbf{A}(r \times k)$, $\mathbf{B}(s \times p)$ and $\mathbf{C}(r \times s)$ are given (known) matrices. We introduce

$$\begin{aligned} \Delta &= \mathbf{A} \hat{\boldsymbol{\theta}} \mathbf{B}^T - \mathbf{C} \\ \mathbf{R} &= n \hat{\boldsymbol{\Sigma}}^* = (\mathbf{Y} - \mathbf{x} \hat{\boldsymbol{\theta}})^T (\mathbf{Y} - \mathbf{x} \hat{\boldsymbol{\theta}}) = \mathbf{Y}^T \mathbf{Y} - \hat{\boldsymbol{\theta}}^T (\mathbf{x}^T \mathbf{x}) \hat{\boldsymbol{\theta}} \end{aligned}$$

and

$$\begin{aligned} \mathbf{E} &= \mathbf{B} \mathbf{R} \mathbf{B}^T \\ \mathbf{H} &= \Delta^T [\mathbf{A} (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{A}^T]^{-1} \Delta. \end{aligned}$$

The likelihood ratio test for testing H_0 against H_1 is equivalent to the test given by the critical region

$$\left\{ \mathbf{y} \mid \frac{\det(\mathbf{e})}{\det(\mathbf{e} + \mathbf{h})} \leq U(s, r, n - k)_\alpha \right\},$$

where $U(s, r, n - k)_\alpha$ is the α quantile in the null-hypothesis distribution of the test statistic (see below).

Question 5.6

Question 5.6

We now want to test if $[\alpha_y \quad \beta_y \quad \gamma_y] = [11 \quad 4 \quad 6]$ against all alternatives. Wilks Lambda for this test is:

Choose one answer

- ☐ 0.453
- ☐ 0.222
- ☐ Don't know
- ☐ 0.004
- ☐ 0.586
- ☐ 0.001

$$A = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Coding:

```
> Delta<-thetahat[,1]-c(11,4,6)
> E<-R[1,1]
> H<-t(Delta)%*%(t(X)%*%X)%*%Delta

> Wilks<-E/(E+H)

> Wilks
      [,1]
[1,] 0.4533333
```

- 0.453 – option 1.

||| Theorem 4.21

We consider the above mentioned situation including the assumption of the normality of the observations. Furthermore we consider the hypothesis

$$H_0 : A \theta B^T = C \quad \text{against} \quad H_1 : A \theta B^T \neq C,$$

where $A(r \times k)$, $B(s \times p)$ and $C(r \times s)$ are given (known) matrices. We introduce

$$\begin{aligned} \Delta &= A \hat{\theta} B^T - C \\ R &= n \hat{\Sigma}^* = (Y - X \hat{\theta})^T (Y - X \hat{\theta}) = Y^T Y - \hat{\theta}^T (X^T X) \hat{\theta} \end{aligned}$$

and

$$\begin{aligned} E &= B R B^T \\ H &= \Delta^T [A (X^T X)^{-1} A^T]^{-1} \Delta. \end{aligned}$$

The likelihood ratio test for testing H_0 against H_1 is equivalent to the test given by the critical region

$$\{y \mid \frac{\det(\mathbf{e})}{\det(\mathbf{e} + \mathbf{h})} \leq U(s, r, n - k)_\alpha\},$$

where $U(s, r, n - k)_\alpha$ is the α quantile in the null-hypothesis distribution of the test statistic (see below).

Exercises

- Exam 2023, Questions 5-7.